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Layered representation diagnostics for short metagenomic reads

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Abstract

Short-read metagenomic next-generation sequencing (mNGS) pipelines process trimmed, ambiguous or locally perturbed reads. Because representation compression precedes classification or database lookup, downstream accuracy alone does not reveal which sequence attributes were retained. We present a representation-diagnostics framework with a dense audit from 50 to 75 bp and anchors at 100, 125 and 150 bp. The resulting canonical 4-mer, property and multi-scale property-pooling representation (CK4P-MSP) combines a reverse-complement canonical 4-mer block (K), a global property block (P) and multi-scale property pooling (MSP) in 222 block-normalized dimensions. A seven-group ablation associated P primarily with global stability, MSP with grouped local-change readability and K with composition-linked retrieval. Across the prespecified whole-genome-sequence perturbation grid, CK4P-MSP had mean paired cosine 0.989, mean drift 0.129 and grouped local-versus-nuisance macro-F1 of 0.979. A shared-template sweep over integer lengths from 50 to 75 bp retained the same operating profile. Across six source-grouped Critical Assessment of Metagenome Interpretation (CAMI) target-versus-background tasks drawn from one source pool, a probe trained on 100-bp clean reads yielded mean shifted macro-F1 of 0.793 for CK4P-MSP. This value was comparable to canonical 4-mer and 5-mer blocks (CK4 and CK5) and descriptively 0.010–0.012 higher than PseKNC and PseEIIP. Train-fitted coordinate scaling exposed an N-mask failure mode. PseKNC achieved lower drift, whereas CK4P-MSP retained stronger grouped local-change readability. Full-position encodings remained higher-dimensional upper bounds for fine positional information, and compressed high-k controls were less stable at the same feature budget. CK4P-MSP therefore occupies an attributable stability–readability trade-off rather than minimizing drift or maximizing predictive accuracy alone. Compact biochemical and coarse positional summaries can provide interpretable representation-level audit coordinates alongside database-linked exact matching and alignment.

Keywords metagenomics, short reads, representation diagnostics, k-mer features, biochemical sequence encoding

Metagenomic next-generation sequencing can interrogate complex or unexpected microbial mixtures without a fixed target panel, and it has become an important route for infectious-disease investigation when targeted assays are insufficient [1, 2, 3]. Yet the reads entering downstream analysis are often short, trimmed, ambiguous and locally corrupted. Adapter removal, quality trimming and low-biomass contamination can all change the effective sequence evidence available to a pipeline before classification begins [4, 5, 6]. Read length can affect homolog recovery and taxonomic assignment, although the magnitude and direction of the effect depend on the classifier, database, taxonomic group and error profile [7, 8, 9, 10]. These constraints do not make short reads intrinsically unusable. Rather, reduced context can narrow the redundancy of local sequence evidence and make it harder to determine which information categories remain available. A read representation may already have compressed local composition, biochemical regularity or positional information before a downstream benchmark reports success or failure. Short-read representation is therefore not merely a preprocessing step: it defines which categories of evidence remain available to a later classifier, alignment routine or database query.

Current metagenomic workflows are strongest when they preserve exact or near-exact match evidence. Canonical k-mer, minimizer, alignment and database-backed systems such as *Kraken*-family tools, *Centrifuge*, *CLARK*, *Kaiju* and related indexing frameworks have established the practical value of explicit sequence matching for taxonomic assignment and downstream biological interpretation [9, 11, 12, 13, 10]. Community benchmarks further show that performance depends on database coverage, sample composition, novelty and evaluation context, not only on the classifier name [14, 15]. These methods should therefore be acknowledged rather than displaced: database-linked match evidence remains the backbone for tasks such as species discrimination, allele-sensitive interpretation and resistance-gene calling. Spaced-seed indexing further extends this line by tuning match patterns for sensitivity under mutation, but its goal is still matching accuracy rather than layered diagnostic analysis [16]. The value of additional audit coordinates can become more visible toward the lower end of the 50–75 bp range, where local matching evidence has less redundancy than at 150 bp. At the same time, these systems are designed to answer a different question from the one asked here. They determine what a read matches, but they do not directly diagnose which kinds of information remain visible in the representation itself when reads are short, noisy or partially degraded.

Vectorized k-mer representations provide the first broad alternative to explicit matching. Here the central operation is to collapse each read or fragment into a composition profile, such as count, frequency, relative-frequency, TF-IDF, or background-adjusted features. PlasFlow and the BMC Bioinformatics 2018 bacteria classifier show that this compact route can be highly competitive when the task is composition-driven rather than order-driven [17, 18]. Resource-saving k-mer classification extends the same logic by showing that low-dimensional k-mer distributions plus classical machine learning can remain competitive even when deep models are not the only option [19]. However, these vectorized k-mer methods rely solely on composition statistics and do not explicitly layer biochemical stability or positional information alongside composition evidence. That distinction becomes more consequential as reads shorten and local composition survives only in compressed form. These methods are not weak baselines; they are a distinct representation family whose success depends on how much local composition survives the short-read regime.

Hashing and sketching form a separate compression family rather than a lower-rank version of singular-value decomposition (SVD). MinHash, locality-sensitive hashing (LSH), histosketch and related strategies map k-mer sets or profiles into fixed-length signatures for rapid similarity search, large-scale database indexing or ultra-lightweight compression [20, 21, 22, 23]. Kssd and the HULK/histosketch line compress representations in service of indexing or neighbourhood search, not in service of preserving linear variance structure. That distinction matters here because our vector route is not asking for the cheapest signature possible; it asks whether a sparse composition vector retains signal accessible to a fixed shallow probe once the short-read regime is held constant. Sketching papers therefore belong in the related work as a neighbouring compression paradigm, not as a substitute for vectorized k-mer analysis.

Biochemical and signal-based encodings add a second alternative representation family. Chaos-game and genomic-signal formulations, together with numerical mappings such as Voss, tetrahedron and electron-ion interaction potential (EIIP), expose sequence structure through algebraic or physicochemical coordinates rather than literal token identity alone [24, 25, 26, 27, 28, 29, 30]. Nucleotide chemical-property encodings have also represented ring structure, chemical functionality and hydrogen-bond class, often together with cumulative nucleotide density or other position-dependent frequencies [31]. These mappings have been used well beyond metagenomic classification, including promoter, regulatory-site, nucleosome-positioning and nucleotide-modification prediction. Their value is therefore not specific to mNGS or to k-mer classifiers. At the same time, the biological interpretation of a named property does not by itself establish that its numerical coordinate is causal, non-redundant or robust across sequence lengths and tasks.

Hybrid handcrafted descriptors already combine composition, physicochemical properties and sequence-order information. Pseudo k-tuple nucleotide composition (PseKNC) augments k-tuple frequencies with lagged correlations between oligonucleotide properties, and PseKNC-General and repDNA expose broad families of autocorrelation, pseudo-composition and user-defined physicochemical descriptors [32, 33, 34]. Pseudo electron-ion interaction potential (PseEIIP) similarly weights trinucleotide composition by EIIP, whereas platforms such as iFeatureOmega catalogue composition, position-specific, autocorrelation and physicochemical encoders within one feature-engineering system [35, 36]. Recent multiscale density encodings further show that local and broader position-dependent nucleotide patterns can be combined with learned models [37]. Thus, the relevant gap is not the absence of prior feature fusion. Most existing hybrid descriptors were developed and selected within task-specific predictive pipelines, where global biochemical summaries and position-dependent components are evaluated as a joint feature set. Their metric-specific contributions, redundancy and failure modes under matched sequence perturbations are less often separated explicitly.

Sequence-token deep learning classifiers make the neighbouring problem space especially relevant. DeepMicrobes introduced deep learning for metagenomic taxonomic classification, using canonical k-mer tokenization, embedding, BiLSTM and attention for read-level assignment [38]. MetaTransformer extended this family with self-attention and alternative tokenization or embedding schemes, including character-level, BPE, k-mer and hash/LSH variants [39]. BERTax showed that a small-k Transformer can support hierarchical taxonomy when pretraining and output heads are aligned to the task [40]. BarcodeBERT further demonstrated that barcode-scale biodiversity tasks benefit from domain-specific pretraining and masking choices, although its setting is barcode classification rather than shotgun short-read mNGS [41]. Large pre-trained DNA language models extend this representation family as general-purpose encoders [42, 43, 44, 45, 46]. Such learned embeddings can provide strong predictive features, particularly when pretraining data and compute are available. Here they define the predictive-performance context for a different question: whether a low-dimensional, training-free and block-decomposable representation can expose local composition, biochemical and coarse positional evidence under controlled short-read perturbation.

Raw-sequence and position-aware models complete the picture. DeepVirFinder, Virtifier, and DETIRE demonstrate that CNN or hybrid architectures can extract useful motif- and composition-level signal from viral fragments, but their target is viral detection rather than multi-class short-read taxonomy [47, 48, 49]. KmerAperture adds an important boundary case by showing that ordinary k-mer set comparisons can lose synteny, whereas matching k-mers back to their original lists can retain positional information about core and accessory differences [50]. Collectively, prior work has established both individual and hybrid sequence encoders. What remains less developed is a common perturbation-audit protocol that decomposes a compact representation into local composition, global biochemical and coarse positional-property blocks, then tests their conditional contributions without treating predictive accuracy as the only endpoint.

Against this background, we treat short-read representation as an audit problem with several distinct endpoints. We ask how local k-mer-composition retention, biochemical perturbation stability, positional readability and compactness trade off across controlled short-read settings. CK4P-MSP combines reverse-complement canonical 4-mer composition with global biochemical summaries and multi-scale property pooling to produce a block-decomposable audit coordinate system. The methodological contribution is the explicit separation of global and positionalized property channels within a matched perturbation protocol, rather than composition–property fusion alone. Full-position matrices serve as positional upper bounds, whereas the canonical spaced-property (CSP) control tests whether matching-oriented spaced-seed priors transfer to dense read-level representations. The study incorporates all seven non-empty K/P/MSP combinations, representative historical handcrafted descriptors, prespecified conditional contrasts, same-dimension reduction controls, paired non-parametric tests, block-weight audits, k-nearest-neighbour estimator-sensitivity checks for empirical separability, dimension-matched high-k compressed k-mer baselines, local mutation-fraction sweeps and a shared-template continuity audit at every integer length from 50 to 75 bp. Broader 100, 125 and 150 bp anchors test whether the lower-range pattern persists without implying a dense sweep over the complete 50–150 bp interval. The aim is to identify each block's metric-specific contribution and operating boundary under a compact, interpretable feature budget, without assuming statistical independence or universal necessity.

Materials and Methods

Study design

We designed the study as a representation-diagnostic analysis of controlled short-read perturbation settings. Here, representation diagnostics denotes a prespecified set of representation-level measurements rather than a clinical decision rule: paired stability, nearest-clean retrieval, grouped local-change readout, feature dimension and blockwise decomposition. The central question was whether biochemical and coarse positional summaries add auditable evidence to a canonical local k-mer-composition backbone. CK4 and CK5 were reverse-complement canonical k-mer-composition baselines; P denoted global biochemical-property summaries; MSP denoted multi-scale positional property pooling. The seven non-empty K/P/MSP combinations were evaluated, full-position matrices were treated as positional upper bounds, and the canonical spaced-property (CSP) control served as a spaced-seed mechanism comparator.

This design makes two questions testable. First, whether a compact mixed representation remains close to its clean counterpart under nuisance perturbation while retaining representation-level readability. Second, whether the observed behaviour can be assigned to metric-specific K, P or MSP contributions rather than to dimensionality, normalization or dilution in a mixed L2 space. We therefore combined a seven-group ablation with a 2×2 spatial-localization-by-substitution-chemistry audit, representative historical handcrafted descriptors, same-dimension principal-component analysis (PCA) and singular-value decomposition (SVD) controls, dimension-matched high-k compressed k-mer baselines, block-weight and property-scaling sensitivity, paired non-parametric tests, k-nearest-neighbour estimator-sensitivity checks for empirical separability, P/MSP relation analysis, feature-extraction runtime and error-aware ART simulator strata. Shallow readout probes tested whether signal remained accessible to a fixed low-capacity model; they were not treated as production classifier performance. Throughout, mixed-space L2 is interpreted as standardized representation drift after block construction and normalization, not as a natural biophysical distance between commensurate physical units.

The lower-range design was anchored to a commonly reported 50–75 bp mNGS read-length regime [51, p. 99]. A shared-template continuity audit evaluated every integer length in this interval by prefix-truncating the same 420 independently sampled 150-bp source windows. This design changed read length without changing source identity. Analyses requiring smaller condition sets retained prespecified subsets of these lower-range lengths for consistency across experimental layers. Broader anchors at 100, 125 and 150 bp tested whether lower-range behaviour persisted as more sequence context became available, and 300 bp was retained only as an extended-read reference in selected boundary analyses. No biological or clinical threshold was assigned to any individual lower-range length. This sampling plan is a dense lower-bound sweep plus broader anchors, not a continuous 50–150 bp scan and not a claim that short reads are unusable.

Representation families

The representation families were defined to separate local k-mer composition, global biochemical summaries and coarse positional information. Canonical 4-mer (CK4) and canonical 5-mer (CK5) blocks count reverse-complement canonical contiguous k-mers. P contains the read-level mean and population standard deviation of hydrogen-bond, GC, purine and electron-ion interaction potential (EIIP) maps, together with N fraction, length divided by 200 and Shannon entropy over A/C/G/T/N divided by $\log_2 5$. MSP mean-pools five per-base maps (hydrogen bond, GC, purine, EIIP and N indicator) over relative-position binsets with 2, 3, 4 and 6 bins. For a read of length L and a scale with b bins, the jth bin spans indices $[\lfloor jL/b \rfloor, \lfloor (j+1)L/b \rfloor)$; any empty bin is represented by zeros. Ambiguous bases contribute zero to the four biochemical maps and one to the N-indicator map. The complete coordinate definitions are reported in Supplementary Table S6. MSP does not create a larger sequence vocabulary or reconstruct full position; it adds a coarse positional property summary to an otherwise order-poor k-mer-frequency vector. The default 2+3+4+6 binset was prespecified as a coarse-to-fine relative-position summary whose finest scale remains above single-position resolution across the 50–150 bp study range. The seven-group ablation comprised K (136 dimensions), P (11), MSP (75), K+P (147), K+MSP (211), P+MSP (86) and K+P+MSP (222). Full-position composition or property matrices retained per-position information and were used only to estimate an upper bound on positional readability. CSP combines spaced-seed counts with property summaries and was used to test whether matching-oriented spaced-seed priors transfer into dense representation diagnostics.

Three historical handcrafted descriptor families were implemented as direct controls. The Type-I PseKNC control used trinucleotide composition ($k = 3$), three lagged correlation terms ($\lambda = 3$), weight $w = 0.05$ and the standardized Rise, Roll,

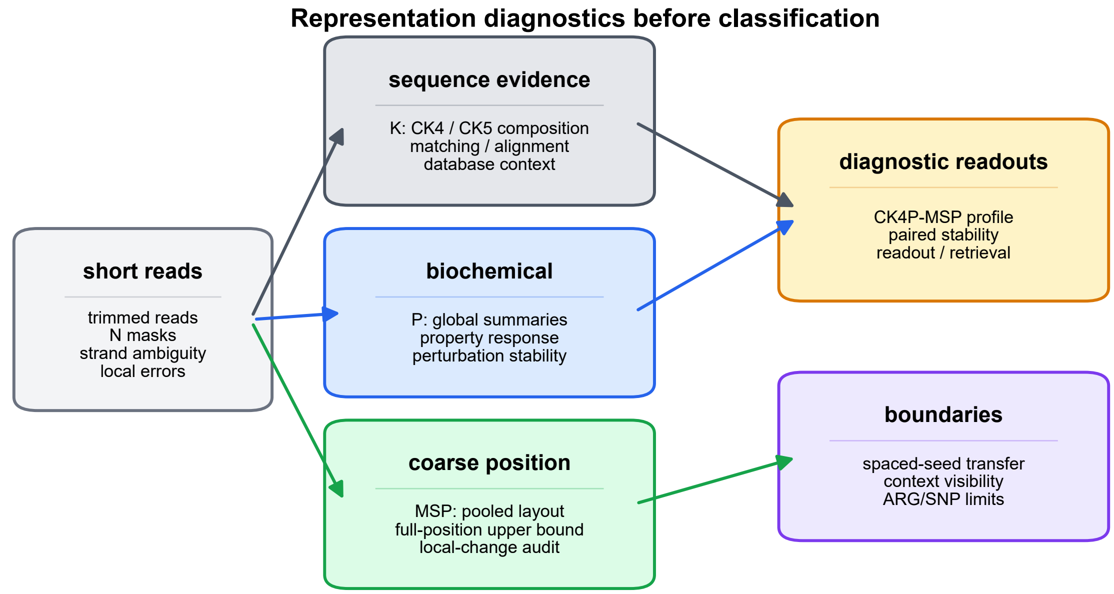

Figure 1 Representation-diagnostic framework and read-length regime. Clean and perturbed reads are encoded through canonical local k-mer composition (K), global biochemical summaries (P) and multi-scale positional property pooling (MSP). The audit reports paired stability, nearest-clean retrieval, grouped local-change readout and feature dimension. Lower-bound behaviour is evaluated densely from 50 to 75 bp, whereas 100, 125 and 150 bp serve as broader anchors; full-position encodings and the canonical spaced-property (CSP) control provide boundary comparisons.

Alt text: Workflow diagram showing short reads separated into local-composition, biochemical and positional representation channels before diagnostic readouts and boundary checks.

Shift, Slide, Tilt and Twist dinucleotide-property tables, yielding 67 coordinates [32, 33]. The nucleotide chemical property plus accumulated nucleotide frequency (NCP+ANF) control interleaved three chemical-property indicators (ring structure, functional group and hydrogen-bond class) with accumulated nucleotide frequency at each position, yielding $4L$ coordinates for a read-length budget L [31]. The 64-dimensional PseEIIP control multiplied each normalized trinucleotide frequency by the sum of its three EIIP values [30, 35]. Because the original descriptors generally assume A/C/G/T input, the audit used declared deterministic extensions for ambiguous reads: invalid PseKNC and PseEIIP windows contributed zero composition mass, PseKNC correlation terms averaged only valid dinucleotide pairs, and N had a zero NCP code plus its own accumulated frequency. Trimmed NCP+ANF reads were N-padded to the declared source-length budget. Every historical matrix was row-normalized before the common drift and readout protocol. These adaptations preserve fixed dimensions and make N-masked conditions evaluable; they are not claimed to be the only possible ambiguous-base convention.

For mixed representations, each block was computed separately and internally L2-normalized before weighted assembly. For read s_i, let $\mathbf{K}_i$, $\mathbf{P}_i$ and $\mathbf{M}_i$ denote the reverse-complement canonical 4-mer composition block, the global biochemical-property block, and the multi-scale positional-property block, respectively. The internally normalized blocks were defined as

$$\hat{\mathbf{K}}_i = \frac{\mathbf{K}_i}{\|\mathbf{K}_i\|_2}, \quad \hat{\mathbf{P}}_i = \frac{\mathbf{P}_i}{\|\mathbf{P}_i\|_2}, \quad \hat{\mathbf{M}}_i = \frac{\mathbf{M}_i}{\|\mathbf{M}_i\|_2},$$

with zero-norm safeguards applied before normalization. The assembled compact representation was then

$$\mathbf{x}_i = \frac{1}{\sqrt{\alpha^2+\beta^2+\gamma^2}} \left[\alpha\hat{\mathbf{K}}_i\,;\,\beta\hat{\mathbf{P}}_i\,;\,\gamma\hat{\mathbf{M}}_i\right],$$

where α, β and γ are block weights and $[\cdot\,;\,\cdot]$ denotes concatenation. Unless otherwise stated, all three weights were set to 1. The block-weight audit compared equal weights with K-dominant $(2,1,1)$ and property-dominant $(1,2,2)$ settings; the separate MSP sensitivity audit evaluated $\gamma \in \{0, 0.25, 0.5, 1, 2\}$. Because the block norms are fixed before concatenation, the denominator is a fixed scalar for a given weight setting rather than a read-specific global norm. The reported global drift metric was

$$d(i,j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2\,,$$

which should be interpreted as standardized representation drift in the assembled feature space, not as a natural physical distance between commensurate biological units. Block normalization fixes the declared contribution of each block but does not make the underlying coordinates physically commensurate. This definition also gives the block-normalized distance identity

$$
\begin{aligned}
d^2_{\mathrm{mix}}(i,j) = \frac{1}{\alpha^2+\beta^2+\gamma^2}\Big[\alpha^2 \left\|\hat{\mathbf{K}}_i-\hat{\mathbf{K}}_j\right\|_2^2 \\
+\beta^2 \left\|\hat{\mathbf{P}}_i-\hat{\mathbf{P}}_j\right\|_2^2 \\
+\gamma^2 \left\|\hat{\mathbf{M}}_i-\hat{\mathbf{M}}_j\right\|_2^2\Big].
\end{aligned}
$$

For comparison with the composition-only drift $d_K(i,j) = \|\hat{\mathbf{K}}_i-\hat{\mathbf{K}}_j\|_2$, the mixed squared drift is lower than d_K^2 when

$$\beta^2\left(d_P^2-d_K^2\right)+\gamma^2\left(d_M^2-d_K^2\right)<0,$$

where d_P and d_M are the corresponding blockwise drifts. Thus, lower property and MSP drift than K drift is a sufficient, but not necessary, condition for lower mixed drift. The diagnostic protocol takes matched clean/perturbed reads, constructs each block without fitting a predictive encoder, reports paired stability and nearest-clean retrieval, tests structured local change with grouped delta-readout, and decomposes the result through the seven-group ablation and prespecified conditional contrasts. This protocol yields a representation profile rather than an automated acceptance threshold. The design therefore does not assume that concatenation is intrinsically useful; it requires the combination to survive dimensionality, weight, counterfactual and block-contribution audits.

After freezing the CK4P-MSP main contract, we evaluated one exploratory extension that couples canonical 4-mer tokens to relative position. The positional k-mer moment (PKM) block accumulates the first three polynomial moments of each canonical 4-mer over normalized read coordinates and compresses the resulting token-position array by deterministic signed hashing to 75 dimensions. The block-normalized extension, termed CK4P-MSP-PKM, appends PKM to K/P/MSP and uses the fixed assembly weight $\delta = 0.25$, for 297 features in total. This value was retained from a prespecified exploratory grid $\delta \in \{0.1, 0.25, 0.5, 0.75, 1\}$ as a supplementary Pareto point rather than as a universally optimal parameter. The complete construction, resampling check and direction-sensitivity audit are reported in Supplementary Figure S15 and Supplementary Methods.

Data layers and perturbation design

Here, WGS denotes whole-genome-sequence-derived templates sampled from public assemblies. ART denotes the ART next-generation sequencing read simulator used for the external error-profile audit.

The experiments were organized into nine main data layers, with the scale of each layer reported in Table 2. The WGS perturbation layer contained 25,200 rows from 3,600 clean templates across six genera, 21 species and six read lengths; all species and NCBI assembly accessions are listed in Supplementary Table S7. Clean windows were sampled from the public assemblies with seed 303 and expanded into clean, 1% substitution, 3% N masking, 5-bp end trimming, combined 1% substitution plus 3% N masking, 1% short-indel and contiguous 6-bp mismatch conditions. The position-property ablation layer expanded this to 36,000 rows over the same template set and length grid. The lower-bound continuity layer generated 76,440 global perturbation rows from 420 shared source templates at all 26 integer lengths from 50 to 75 bp, seven K/P/MSP representations and six non-clean perturbations. Its grouped local audit generated 18,720 matched template triplets, or 56,160 clean/local/nuisance variants, from 180 triplets per length-by-mode cell across four local modes. ART Illumina-like simulation contributed 55,961 matched clean/simulated pairs, represented as 111,922 rows, across 50, 60, 69, 75, 100, 125 and 150 bp [52]. Single-end reads were generated with the ART Illumina executable's default error profile, 0.005-fold coverage, SAM output and a base seed of 606 offset by read length; "paired" in this manuscript refers to each simulated read and its aligned clean template, not paired-end sequencing. The current-contract Critical Assessment of Metagenome Interpretation (CAMI) TOY low-complexity layer contained 16,342 eligible source-read groups after deterministic mate collapse and length filtering [14]. Six prespecified target-versus-background tasks were expanded to 129,600 task-specific rows at 69, 75 and 100 bp under clean, 3% N masking and 1% substitution conditions; source reads could recur as background across tasks, so these rows are not unique biological observations. The CAMI II marine lightweight probe used 4,000 anonymous source reads and 48,000 length/condition rows at 69, 75 and 100 bp [15]. The main local mutation layer used 250 triplets per analysis cell, four local modes and three read lengths. WGS-derived paired perturbations measured clean-versus-perturbed stability; the continuity layer isolated lower-range length effects on shared templates; ART added simulator-derived errors; CAMI resources supplied bounded external probes; controlled position and order tasks tested positional readability; and boundary probes tested spaced-seed transfer, context visibility and ARG/SNP limits. Experiment-specific seeds and complete argument sets are retained in the machine-readable run manifests.

The supplementary CK4P-MSP-PKM audit additionally used the public Dorsal binding-site and cis-regulatory-module sequences released by Clifford and Adami [53]. Exact known-site occurrences were located within their corresponding regulatory sequences, and 1,125 windows of 50, 75, 100 or 150 bp were labelled by whether the site occurred in the left, middle or right third of the window. The 32 source regulatory modules defined the cross-validation groups, so windows from one module did not cross folds. This motif-position probe tests access to a bounded position label; it is not a transcription-factor-binding-site discovery, affinity or regulatory-function benchmark.

For CAMI_TOY_low, read identifiers were joined to the available taxon mapping and normalized to a common NCBI taxon-identifier field. One deterministic representative was retained per `clean_read_id`; sequences shorter than 100 bp were excluded. The six most abundant eligible labels were selected before model fitting. They comprised five species-level taxa and one strain-level taxon; their NCBI Taxonomy identifiers, scientific names and ranks are reported in Supplementary Table S9 [54]. Each binary task used 1,200 target source groups and 1,200 background groups sampled across the remaining labels. Every source was then truncated to 69, 75 and 100 bp and expanded into clean, `N_3pct` and `substitution_1pct` conditions. The resulting 21,600 rows per task were evaluated by fivefold `StratifiedGroupKFold`, with all length and perturbation derivatives of a source retained in one fold. These source-grouped tasks measure coarse target-versus-background transfer and do not estimate fine taxonomic classification.

Table 1. Representation families and their prespecified diagnostic roles. The rows distinguish compact K/P/MSP block combinations, historical handcrafted descriptors, full-position upper bounds and the spaced-seed mechanism comparator. Interpretations specify the evidence assigned to each family and do not constitute a single-metric ranking.

family	information channel	main role	main-text interpretation
CK4 / CK5	canonical local k-mer composition	composition backbone and comparator	compact token-frequency evidence, not database-linked read identity
P	global biochemical summaries	perturbation-stable channel	tests global biochemical response without local composition
MSP	positionalized biochemical summaries	coarse positional channel	tests local-change readability without K or global P
CK4+P	composition plus global biochemical signal	global property baseline	tests whether biochemical summaries add to CK4
CK4+MSP	composition plus positionalized biochemical signal	positional property ablation	tests whether MSP returns coarse layout information without global P
P+MSP	global plus positionalized biochemical summaries	property-only combination	tests property stability and retrieval without K
CK4P-MSP	composition plus multi-scale biochemical layout	compact main representation	balances stability, readability and composition-linked retrieval
PseKNC / NCP+ANF / PseEIIP	historical composition–property, order or position descriptors	direct handcrafted-feature controls	define whether CK4P-MSP offers a distinct attributable trade-off rather than first feature fusion
full-position matrix	per-position token or property signal	upper-bound diagnostic	tests fine positional information at higher dimensional cost
CSP	spaced-seed prior plus biochemical summary	spaced-seed transfer comparator	delimits where matching-oriented spaced seeds transfer to dense features

We also added a lightweight CAMI II marine subset as an external short-read stability probe. We streamed a prefix of CAMI II marine short-read sample 0 and parsed 4,000 anonymous 150 bp reads from the embedded `anonymous_reads.fq.gz` member. These reads were truncated to 69, 75 and 100 bp and expanded into clean, `N_3pct`, `substitution_1pct` and `substitution_1pct_N_3pct` conditions, yielding 48,000 length/condition rows. CAMI II provides benchmark truth resources, but the anonymous FASTQ headers did not provide read-level taxonomic labels in the streamed-read file used here, and the separate OTU-specific truth bundles were not reconstructed for this lightweight analysis. This CAMI II probe therefore reports paired perturbation-stability metrics on an external metagenomic sequence source rather than CAMI II taxonomic validation.

This layered simulation design was used because each component isolates a different failure mode. The controlled WGS grid makes perturbation type, read length and representation family directly comparable. The continuity layer isolates the lower read-length variable through shared-template truncation; its dense points are descriptive sensitivity evidence and do not multiply the prespecified inferential grid. ART adds a recognized Illumina-like simulator layer under the current CK4P-MSP contract. Because the ART profile produces non-monotonic cycle-dependent error loads, it supports within-length representation comparisons and quality-stratified ordering rather than a causal claim that every change across 50–150 bp is due to length alone.

Metrics and statistical analysis

Perturbation stability was quantified by paired cosine similarity, paired L2 drift and nearest-clean retrieval. Retrieval searched at most 500 clean candidates within each matched analysis cell; its random-candidate chance level was therefore approximately $1/n_{\mathrm{pool}}$, and saturation at 1.0 limits its ability to rank strong K-containing representations. Representation readability was quantified by macro-F1 and accuracy from shallow logistic-regression or nearest-centroid probes. Except for the explicitly defined CAMI contract-space audit, logistic probes used training-partition-standardized inputs, scikit-learn's default L2 penalty and $C = 1$, a fixed random seed, and a 70:30 stratified split for global readouts; local-change probes used fivefold stratified grouped cross-validation. The maximum iteration count was 5,000 for manuscript readouts and 4,000 in the dimensional-reduction audit. These readout metrics were accessibility probes for low-dimensional features, not production classifier performance. Compactness was measured by feature dimension. Local mutation analysis used selective-sensitivity ratios and grouped delta-readout; all derivatives of one template were kept in the same fold through template identifiers. A selective-sensitivity ratio was treated as undefined when the matched nuisance drift was at or below 10^{-8}; valid-ratio counts and fractions were retained so that near-zero denominators could not produce arbitrarily large finite ratios. For the seven-group audit, three contrasts were prespecified: K+P+MSP versus P+MSP for the conditional K contribution, versus K+MSP for P, and versus K+P for MSP. Mean matched-cell differences were summarized by 10,000 paired bootstrap resamples. Two-sided paired Wilcoxon signed-rank tests were adjusted across the 12 metric-by-contrast rows by the Benjamini–Hochberg procedure [55, 56]. The analysis unit for these tests was the matched length-by-perturbation or length-by-local-mode cell, not an independent biological cohort. Accordingly, bootstrap intervals describe variation across the prespecified analysis grid and are not population-level confidence intervals for biological cohorts. Same-dimension PCA/SVD controls used randomized PCA or TruncatedSVD fitted on the training partition at 147- and 222-feature targets with the experiment seed; block-weight audits tested reasonable reweightings of the composition and property blocks.

Table 2. Data layers, analysis scales, diagnostic questions, primary metrics and claim boundaries. Row counts include perturbation and task derivatives where stated and therefore are not counts of independent biological observations.

data layer	scale	diagnostic question	primary metrics	claim boundary
WGS perturbation pairs	25,200 rows; 3,600 clean templates; 6 genera/21 species; lengths 69,75,100,125,150,300 bp	Which representations keep clean and perturbed reads close?	paired cosine, L2 drift, retrieval	controlled representation stability
Position-property ablation	36,000 rows; 3,600 templates; 6 genera/21 species; lengths 69,75,100,125,150,300 bp	What do P and MSP add to the CK4 composition backbone?	paired stability, block drift, delta-readout, MI proxies	decomposable representation audit, not a natural physical metric
Shared-template length continuity	420 shared templates at every integer length from 50 to 75 bp; 76,440 global rows and 18,720 local triplets	Does the stability–readability profile persist across the lower short-read range?	paired drift, grouped delta-readout, conditional P/MSP increments	descriptive lower-bound sensitivity; not a dense 50–150 bp scan
ART Illumina-like simulation	55,961 matched clean/simulated pairs; 111,922 rows; lengths 50,60,69,75,100,125,150 bp	Does current-contract ordering persist within simulator lengths and quality strata?	drift ratio versus CK4, observed base difference rate	within-length simulator context; not a causal length trend or clinical error model
CAMI_TOY_low fixed-head probes	16,342 eligible source groups; 6 binary target tasks; 2,400 groups and 21,600 task rows per target; lengths 69,75,100 bp	Is coarse label information retained under length and perturbation shifts without refitting?	fixed-head macro-F1, retention, MCC, AUROC/AUPRC	source-grouped coarse-label transfer; not fine taxonomic classification
CAMI II marine anonymous-read probe	48,000 rows from 4,000 anonymous source reads; lengths 69,75,100 bp	Do stability trends persist in a more complex external metagenomic read source?	paired cosine, L2 drift, retrieval	paired stability without reconstructed read-level taxonomic labels
controlled position tasks	controlled synthetic tasks over the short-read length grid	What does fine positional resolution add?	macro-F1, feature dimension	upper-bound diagnostic
local mutation sensitivity	250 triplets per analysis cell; 12 representations; 4 local modes; lengths 69,100,150 bp	Can local biochemical change be detected beyond equal-count noise?	sensitivity ratio, delta-readout macro-F1	selective sensitivity, not functional effect prediction
boundary probes	spaced-seed, context-visibility and ARG/SNP boundary screens	Where do spaced-seed, context and ARG/SNP claims stop?	seed stability, visibility threshold, macro-F1	mechanism boundary and failure modes

The CAMI fixed-head audit used the representation contract directly, without coordinate-wise variance scaling. Within each fold, a class-balanced L2-penalized logistic probe with $C = 1$ was fitted only to 100-bp clean training sources and then reused without refitting on every held-out length-by-condition derivative. Macro-F1, balanced accuracy, Matthews correlation coefficient, area under the receiver operating characteristic curve and area under the precision–recall curve were recorded; macro-F1 was the prespecified display metric. Source-stratified bootstrap intervals retained paired predictions within each binary task. Cross-target summaries treated the six target definitions as descriptive blocks. Target-bootstrap intervals and paired signed-rank summaries were retained to characterize consistency across those definitions, but the tasks share one CAMI source pool and can reuse source reads as background. They therefore do not provide independent-community inference. Logistic $C \in \{0.1, 1, 10\}$ was examined as a sensitivity analysis; $C = 1$ remained the prespecified result, and no value was selected from test performance.

The mixed-feature question was also tested through empirical mutual-information (MI) audits. For a perturbation label Y and representation-derived distance features X, an equal-frequency discretized screen was retained as a supplementary descriptive check. The primary analysis used a Ross/Kraskov–Stögbauer–Grassberger (KSG)-style nearest-neighbour estimator with $k = 5$ on low-dimensional continuous summaries and a discrete local-versus-nuisance label [57, 58]. It evaluated d_K, $d_K + d_P$, $d_K + d_M$ and $d_K + d_P + d_M$ across 12 local-mutation cells, with 100 shared label permutations and 100 stratified subsamples per cell. Signed joint-minus-K increments were summarized by a 10,000-resample paired-cell bootstrap and a paired Wilcoxon test; the aggregate permutation P value was calculated from the mean signed increment under the shared null permutations. These estimates are estimator-dependent empirical summaries rather than absolute physical information quantities. The seven-group contribution audit was paired with a relation audit based on rank, canonical-correlation analysis (CCA) and row-wise P/MSP alignment. Together, these analyses test conditional empirical roles; they do not imply that P and MSP measure unrelated physical quantities.

The mechanism-aligned local-change audit crossed spatial organization (contiguous versus dispersed substitutions) with substitution chemistry (property-directed versus uniformly random alternatives). It used the same seven K/P/MSP combinations and template-grouped splits. Two readouts were prespecified: spatial localization (contiguous versus dispersed) and substitution chemistry (property-directed versus random). A separate scaling audit recomputed P and MSP under the declared mappings, fixed per-property $[0, 1]$ maps and train-fit coordinate z-scores. Leave-one-property-family-out variants tested whether one P family alone determined the stability result. These analyses distinguish the intended block roles from a universal claim that every block improves every metric.

We added a dimension-matched high-k compression audit to avoid comparing CK4P-MSP only with short contiguous k-mer baselines. Canonical $k = 15$ signals were compressed to the 222-feature CK4P-MSP budget using hashing-trick counts and sparse random projection. These vector controls were evaluated on the same 69, 75, 100 and 150 bp perturbation grid using paired cosine similarity, L2 drift, nearest-clean retrieval and shallow readout. MinHash was assessed separately in its native estimated-versus-exact Jaccard geometry and was not treated as an L2 vector or linear-readout feature. This audit tests whether compact stability persists against high-specificity k-mer information under a comparable vector budget. The historical-descriptor audit used 30 matched WGS length-by-perturbation cells (five lengths and six perturbations) and 12 template-grouped local-change cells (three lengths and four local modes). Mean differences from CK4P-MSP were summarized by 10,000 paired-cell bootstrap resamples and two-sided paired Wilcoxon tests, with Benjamini–Hochberg adjustment across the eight historical-control contrasts. Feature-extraction runtime used identical, actual batches of 10,000 clean 75-bp reads for every method, with one untimed 100-read warm-up, randomized method order within each of five repeats, garbage collection before timing and one numerical-library thread. Because the clean source panel was smaller than the timing batches, one seeded resampling with replacement was generated for each batch size before timing and then reused across methods. A separate actual 100,000-read batch was timed once per method as a scaling check. Wall-clock time was measured with `perf_counter`; the 10,000-read measurements were summarized by the median and interquartile range. Measurements were made on an Intel Core Ultra 5 125H workstation with 31.6 GB RAM under Python 3.13.9, NumPy 2.3.5 and scikit-learn 1.7.2. These timings describe the tested Python implementations rather than optimized algorithmic limits.

Reproducibility

The reference implementation and contract-v2 result namespace are maintained in the release repository. The manuscript-facing API is `methods/ck4p_msp.py`; historical descriptor definitions are centralized in `methods/historical_descriptors.py`, and their common audit is executed by `experiments/audits/run_historical_descriptor_audit.py`. The contribution, high-k, KSG, factorial, property-scaling, MSP-sensitivity and redundancy/runtime audits are likewise executed from `experiments/audits/` and mapped to their result tables in `docs/contract_v2_evidence_map.md`. Historical scripts and outputs remain available for provenance but are not aliases for the declared block-normalized CK4P-MSP method. Analyses used Python 3.13.9, NumPy 2.3.5, pandas 2.3.3, SciPy 1.16.3, scikit-learn 1.7.2 and Matplotlib 3.10.6 on Windows 11 (build 26200), an Intel Core Ultra 5 125H processor (14 cores/18 threads) and 31.6 GiB RAM. Runtime measurements were single-process wall-clock timings after one untimed warm-up and five timed repetitions. Peak Python-managed allocation was measured in a separate pass with `tracemalloc`; this excludes memory retained solely by native-library allocators. Both are machine-specific engineering measurements rather than hardware-independent complexity estimates.

Results

Empirical mixed-feature information audit under local perturbation

We first tested whether the added property channels showed perturbation-associated signal rather than only changing the scale of a mixed feature space. The contract-v2 KSG-style audit was applied to low-dimensional blockwise distance summaries for local change versus matched nuisance perturbation. Across 12 local-mutation cells, the mean estimate was 0.889 bits for $d_K + d_P + d_M$ and 0.752 bits for d_K alone. The mean signed empirical separability increment was 0.138 bits (paired-cell bootstrap 95% CI 0.073–0.207; paired Wilcoxon $P = 4.88 \times 10^{-4}$; aggregate label-permutation $P = 0.0099$). The increment varied across the tested lengths and attenuated at 150 bp, so it should not be read as length-invariant. Under the tested perturbation grid and estimator, P/MSP distance summaries therefore improved local-versus-nuisance separability beyond the CK4 distance summary.

This audit complements the ablation results: the property blocks contribute empirically detectable perturbation signal under the tested local-change regime, while the block-normalized metric remains standardized representation drift rather than a physical distance with commensurate units.

Estimator sensitivity supports this result (Supplementary Figure S2 and Supplementary Table S3). A separate same-dimension counterfactual tested whether the result followed from adding coordinates with similar marginal scales. Across 20 WGS stability cells, sequence-linked CK4P-MSP had mean drift 0.135, compared with 0.160 after jointly permuting the P/MSP rows and 0.629 for marginally matched Gaussian controls. Across 12 grouped local-change cells, its logistic macro-F1 was 0.979, compared with 0.879 and 0.886 for the permuted and Gaussian controls, respectively. The controls therefore preserved dimensional expansion without preserving the observed sequence-to-property mapping (Supplementary Figure S5).

Global and positional property layers have related but distinct roles

We then evaluated all seven non-empty K/P/MSP combinations over 69, 75, 100 and 150 bp reads. Pure property summaries had the lowest numerical drift (P, 0.025; MSP, 0.027; P+MSP, 0.027), but they did not preserve the same nearest-clean behaviour as K-containing representations: retrieval was 0.295 for P and 0.942 for P+MSP, compared with 1.000 for CK4P-MSP. The composition block therefore traded some numerical stability for composition-linked retrieval. Relative to P+MSP, adding K increased retrieval by 0.058 (95% bootstrap CI 0.036–0.081; BH-adjusted $q = 7.3 \times 10^{-4}$) while increasing, rather than decreasing, drift.

Conditional effects differed by audit axis. Adding P to CK4+MSP reduced mean drift from 0.156 to 0.129, a matched-cell improvement of 0.027 (95% CI 0.023–0.031; $q = 2.4 \times 10^{-7}$), but did not materially change grouped delta-readout (0.9780 versus 0.9786; $q = 0.525$). Adding MSP to CK4+P produced a similar drift reduction of 0.027 (95% CI 0.023–0.032; $q = 2.4 \times 10^{-7}$) and increased grouped local-change macro-F1 from 0.904 to 0.979 (mean difference 0.074, 95% CI 0.035–0.117; $q = 7.3 \times 10^{-4}$). Adding K to P+MSP increased macro-F1 by only 0.008 and the interval included zero. These results assign P primarily to global stability, MSP to local-change readability and K to composition-linked retrieval under the tested grid; they do not establish statistical independence or universal necessity.

The redundancy audit also cautions against overinterpreting P and MSP as separable sources. Across the tested lengths, the first canonical correlation between P and MSP was near one and their first principal components were strongly aligned, as expected because both blocks are derived from the same biochemical property family. However, MSP had higher numerical rank than P, and row-wise P/MSP alignment after compression was incomplete. We therefore interpret the blocks as related layers at different scales, not as interchangeable evidence sources or as measurements of unrelated physical quantities.

The 2×2 mechanism audit made that distinction more specific. Conditional on CK4+P, adding MSP increased spatial-localization macro-F1 by 0.065 (95% CI 0.034–0.098; BH-adjusted $q = 9.77 \times 10^{-4}$) and substitution-chemistry macro-F1 by 0.092 (95% CI 0.059–0.130; $q = 9.77 \times 10^{-4}$). Conditional on CK4+MSP, adding P did not improve spatial localization (difference < 0.001; $q = 0.520$) but increased chemistry readout by 0.030 (95% CI 0.013–0.048; $q = 0.0315$). Thus, MSP contributed coarse localization and chemistry-readable structure, whereas P contributed a smaller global chemistry increment without a detectable spatial increment. The blocks were empirically complementary on selected axes, not statistically independent or individually necessary for every task (Supplementary Figure S11 and Supplementary Table S4).

The property-scaling audit further bounded the geometric interpretation. CK4P-MSP drift was 0.128 under the declared maps and 0.132 under fixed per-property $[0, 1]$ maps, while train-fit coordinate z-scoring increased drift to 0.390. Leave-one-property-family-out P variants had a narrow drift range of 0.155–0.156. The stability pattern is therefore insensitive to reasonable fixed remapping and is not determined by one P coordinate family, but it is not invariant to arbitrary variance amplification. P and MSP should accordingly be described as encoding-value-weighted summaries (Supplementary Figure S11 and Supplementary Table S5).

The P/MSP contribution and relation audits are shown in Supplementary Figures S8 and S9; the mechanism and scaling audits are shown in Supplementary Figure S11. A common-protocol implementation-time comparison with historical descriptors is reported in Supplementary Figure S12.

MSP short-bin reliability persists across 50–75 bp

We next tested whether MSP becomes noise-dominated when short reads are divided into finer relative-position bins. The shared-template continuity audit did not show an abrupt failure across the 26 integer lengths from 50 to 75 bp. At 50 bp, mean grouped local delta-readout macro-F1 was 0.961 with the coarsest 2-bin summary and 0.988 with the complete 2+3+4+6 binset; the complete binset remained between 0.978 and 0.999 across the lower-range grid. Thus, finer relative bins increased rather than erased the structured local-change signal under this controlled construction.

The same audit separated the conditional roles of P and MSP. Across 50–75 bp, adding P to K+MSP reduced mean paired drift by approximately 0.029–0.036, whereas adding MSP to K+P increased grouped local-change macro-F1 by approximately 0.011–0.043, depending on length. CK4P-MSP drift decreased smoothly from 0.174 at 50 bp to 0.139 at 75 bp, while CK4 decreased from 0.294 to 0.235; grouped local-change macro-F1 was 0.988 and 0.993 for CK4P-MSP at the two endpoints. These curves support persistence of the stability–readability profile over the lower-range sweep, not a universal monotonic law for arbitrary sequencing data.

The two-length weight scan at 75 and 100 bp likewise showed that $\gamma = 1$ was not a fragile point. Increasing γ to 2 further reduced global drift, so the default remains a fixed balanced setting rather than an empirically optimized stability weight. Static MSP is therefore usable in the tested short-read regime, but it remains a coarse positional summary rather than an optimized variance-aware estimator. MSP binset and weight sensitivity are shown in Supplementary Figure S6, and the full 50–75 bp continuity audit is shown in Supplementary Figure S13.

Error-aware ART perturbation audit

The current-contract ART audit evaluated CK4, CK4+P, CK4+MSP, CK4P-MSP and CK5 at 50, 60, 69, 75, 100, 125 and 150 bp. Within every length, CK4P-MSP retained 0.58–0.60 of the CK4 standardized drift, and CK4+P and CK4+MSP each retained approximately 0.70–0.73. The ordering was also stable within low-, middle- and high-quality strata, indicating that the property-aware reduction was not created by pooling all simulator-derived reads into one error group.

The observed same-coordinate base-difference rate was not monotonic in read length: it rose from 2.70% at 50 bp to 6.64% at 100 bp, then fell below 0.22% at 125 and 150 bp under the selected ART profile. Consequently, the ART data are used for within-length and within-stratum representation comparisons, not to infer a physical read-length response across independently generated simulator cycles. ART also remains a simulator rather than a complete model of clinical sequencing error.

Quality-stratified results and the observed ART error-load profile are shown in Supplementary Figure S3.

Compact biochemical summaries reduce perturbation drift

The main compact-representation audit tested whether property-aware summaries remained closer to their clean counterparts than composition-only or compressed high-k baselines. Across the WGS-derived perturbation grid, CK4P-MSP retained paired cosine of 0.989 and mean L2 drift of 0.129 at 222 features, compared with 0.969 and 0.220 for CK4 (136 features) and 0.945 and 0.295 for CK5 (512 features). CK4+P retained paired cosine of 0.984 and drift of 0.157 at 147 features. Its conditional comparison with CK4+MSP, rather than an obsolete implementation alias, supports the role of P in global stability.

Dimension-matched high-k vector controls further addressed whether a conventional high-specificity k-mer representation could show the same behaviour under the CK4P-MSP feature budget. At approximately 222 features, CK4P-MSP retained paired cosine 0.989 and L2 drift 0.129, compared with 0.821 and 0.524 for hashed $k = 15$ counts and 0.868 and 0.447 for sparse random projection of $k = 15$ counts. Shallow-readout differences were modest and did not favour CK4P-MSP over all high-k controls, so the result supports a compact perturbation-stability advantage over these training-free compressed vectors

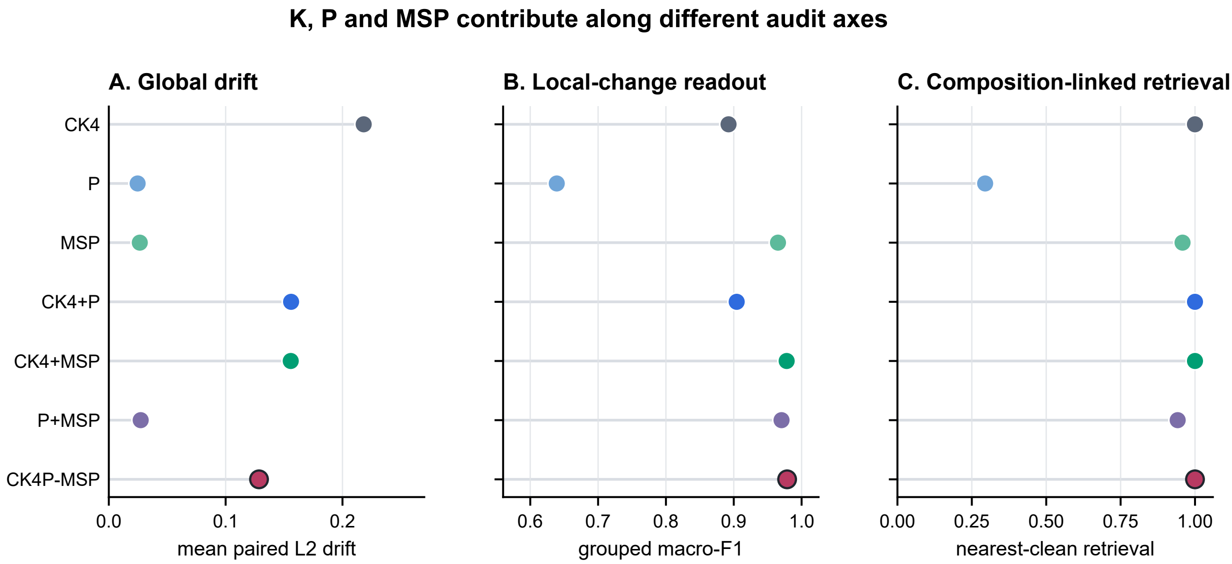

Figure 2 Seven-group K/P/MSP ablation across 24 matched length-by-perturbation cells and 12 grouped local-mutation cells. (A) Mean paired L2 drift is lowest for property-only summaries; CK4P-MSP is more stable than CK4 while retaining the K block. (B) MSP-containing representations preserve the strongest grouped local-versus-nuisance delta-readout under template-grouped fivefold cross-validation. (C) K-containing representations retain near-perfect nearest-clean retrieval from pools of at most 500 candidates, whereas P-only and P+MSP lose composition-linked retrieval. Symbols show means across cells; exact conditional contrasts, 95% paired-cell bootstrap intervals, two-sided Wilcoxon tests and Benjamini–Hochberg-adjusted q values are reported in Supplementary Table S2. These cell-level intervals and tests quantify consistency across the controlled grid, not independent biological-cohort inference.

Alt text: Three aligned dot plots compare all seven K/P/MSP block combinations by global paired L2 drift, grouped local-change macro-F1 and nearest-clean retrieval. Property-only summaries have the lowest drift, MSP-containing summaries have the strongest local-change readout, and K-containing summaries have the strongest retrieval.

rather than a predictive-performance claim. MinHash was assessed separately through its native estimated-versus-exact Jaccard agreement (mean estimated Jaccard 0.711, exact Jaccard 0.711, mean absolute error 0.021); it was not used as an L2-vector baseline. Paired Wilcoxon tests over matched analysis cells supported the primary stability contrasts after multiple-testing correction.

Training-fitted reductions defined a separate boundary. CK7 PCA/SVD projections fitted to the clean partition at 147 or 222 target dimensions achieved mean drift of 0.093–0.098, lower than CK4P-MSP, while their logistic readout remained task-limited (macro-F1 0.403–0.416; Supplementary Figure S1). Thus, learned linear projection can optimize numerical stability more strongly than the fixed biochemical construction. CK4P-MSP's distinguishing property is not universal minimum drift; it is a training-free, block-decomposable representation whose K, P and MSP contributions remain directly attributable.

The fitted-reduction boundary and fixed-contract block-weight audit are summarized in Supplementary Figure S1.

Compressed high-k and native MinHash results are provided in Supplementary Figure S7.

Historical descriptors define a distinct stability–readability boundary

Because composition–property fusion predates CK4P-MSP, we compared the proposed block decomposition with representative PseKNC, NCP+ANF and PseEIIP descriptors under the same paired protocol. Across 30 WGS length-by-perturbation cells, 67-dimensional PseKNC achieved the lowest mean standardized drift (0.113), compared with 0.126 for CK4P-MSP. The paired difference favoured PseKNC by 0.0133 (95% CI 0.0103–0.0163; BH-adjusted $q = 4.97 \times 10^{-8}$). CK4P-MSP therefore did not minimize either drift or dimensionality. It nevertheless had lower drift than CK4, NCP+ANF and PseEIIP by 0.0881, 0.1021 and 0.0316, respectively; all three paired intervals excluded zero after multiplicity adjustment.

The ordering reversed for structured local-change readability. Across 12 template-grouped length-by-local-mode cells, CK4P-MSP reached mean logistic macro-F1 of 0.979, compared with 0.892 for CK4, 0.887 for PseKNC, 0.959 for NCP+ANF and 0.870 for PseEIIP. The paired CK4P-MSP increments were 0.086 (95% CI 0.042–0.134; $q = 5.58 \times 10^{-4}$), 0.092 (0.062–0.124; $q = 5.58 \times 10^{-4}$), 0.020 (0.005–0.036; $q = 0.0161$) and 0.109 (0.074–0.145; $q = 5.58 \times 10^{-4}$), respectively. By contrast, mean global WGS logistic readout differed only modestly (0.400 for CK4P-MSP, 0.396 for PseKNC and 0.394 for PseEIIP), and does not support a predictive-superiority claim.

On actual 10,000-read batches, the optimized reference implementation required a median 2.71 s per 10,000 75-bp reads (interquartile range 1.51–2.86 s), compared with 2.15 s for CK4, 58.01 s for the current PseKNC implementation, 1.09 s for NCP+ANF and 0.89 s for PseEIIP. The separate 100,000-read pass required 28.72 s for CK4P-MSP and 21.73 s for CK4; the corresponding per-10,000-read times (2.87 and 2.17 s) were close to the 10,000-read medians. Runtime is implementation-specific, but the optimized CK4P-MSP path is close to the CK4 reference rather than carrying the earlier repeated-scan overhead. Taken together, the historical audit identifies a Pareto-like boundary: PseKNC is smaller and more

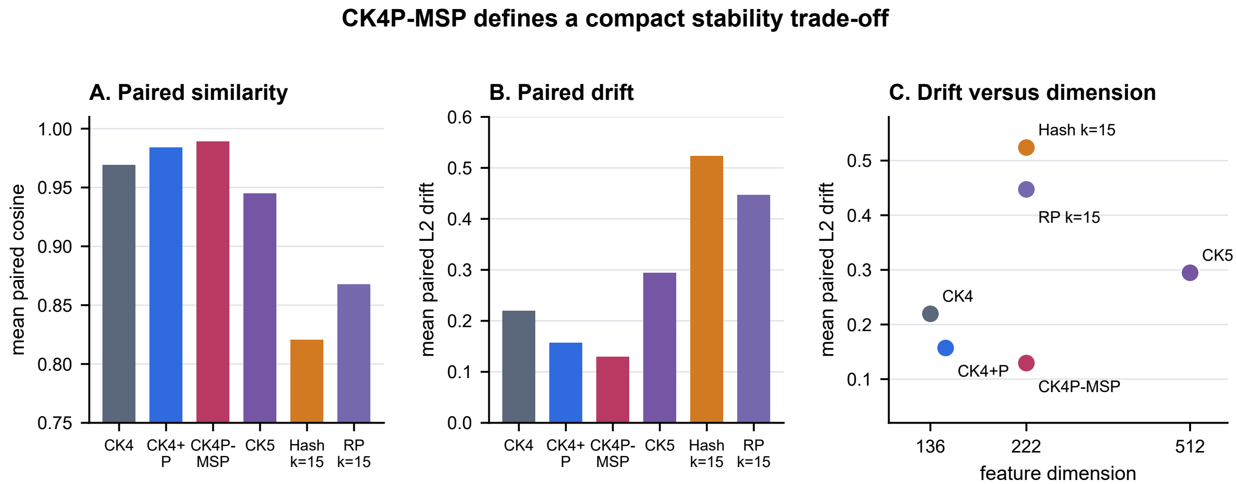

Figure 3 Dimension-matched compact stability audit over 24 length-by-perturbation cells from the controlled WGS grid. Panels compare mean paired cosine similarity, standardized L2 drift and feature dimension across compact composition/property representations and 222-dimensional hashed or sparse-random-projection $k = 15$ vectors. Bars and points show means across matched cells. CK4P-MSP is more stable than CK4, CK5 and both compressed high-k vector controls; lower drift and higher cosine indicate greater paired stability. MinHash is excluded from this L2-vector comparison and evaluated in native Jaccard geometry (Supplementary Figure S7).

Alt text: Bar plots and a scatter plot comparing compact k-mer, property-aware and dimension-matched high-k vector representations by paired similarity, standardized representation drift and feature dimension.

invariant, whereas CK4P-MSP preserves a stronger, explicitly block-attributable response to structured local change (Supplementary Figure S12). The result distinguishes an operating profile; it does not establish first use of composition–property fusion or universal superiority.

CK4P-MSP balances compactness, stability and representation readability

We next evaluated CK4P-MSP as a compact trade-off rather than as a single-metric winner. The seven-group result showed why the complete representation contains all three blocks: property-only variants were numerically stable but weakened nearest-clean retrieval, P improved global stability after conditioning on K+MSP, and MSP improved local-change readout after conditioning on K+P. The historical comparison further showed that lower drift alone can be obtained with PseKNC, but at lower structured local-change readability. CK4P-MSP therefore preserves a compact canonical local-composition backbone while exposing global biochemical and coarse positional audit channels whose contributions can be inspected separately.

The resulting trade-off supported the intended role. CK4P-MSP remained compact, reduced perturbation drift relative to CK4 and CK5 composition baselines, retained near-perfect nearest-clean retrieval and preserved accessible local-change signal. It is neither the minimum-drift historical descriptor nor the full-position readability upper bound. Its role is to provide a block-decomposable audit coordinate system between composition-only compact features, smoother hybrid descriptors and high-dimensional full-position matrices, rather than to maximize predictive accuracy against learned embeddings.

External ART and CAMI probes separate stability from fixed-head label retention

ART and CAMI were used to test whether the representation family showed consistent behaviour outside the controlled WGS grid while keeping stability and label retention as separate questions. Across the seven current-contract ART lengths, CK4P-MSP retained 0.58–0.60 of CK4 drift, while CK4+P and CK4+MSP retained approximately 0.70–0.73. This within-length ratio is emphasized because the simulator error load varied non-monotonically across cycle profiles. CAMI II marine supplied anonymous-read stability. Across nine CAMI II length-by-perturbation cells, mean drift was 0.144 for CK4P-MSP, compared with 0.175 for CK4+P, 0.243 for CK4 and 0.325 for CK5.

CAMI_TOY_low supplied six labelled fixed-head tasks. Averaged over six targets and eight shifted length-by-condition cells per target, mean macro-F1 was 0.7934 for CK4P-MSP, 0.7935 for CK4, 0.7945 for CK4+P, 0.7933 for CK4+MSP and 0.7948 for CK5. The compact K/P/MSP variants therefore had comparable coarse-label accessibility at this resolution rather than a meaningful predictive ranking. CK4P-MSP was descriptively 0.0102 macro-F1 above PseKNC and 0.0120 above PseEIIP, with the same direction in all six target definitions. Because these tasks reused one source pool and were not independent-community replicates, the target-bootstrap intervals and signed-rank calculations are sensitivity summaries rather than population-level inference. Mean retention relative to each method's own 100-bp clean baseline was 0.971 for CK4P-MSP and 0.963 for CK4, whereas PseKNC and CK5 reached 0.974 and 0.972, respectively. CK4P-MSP thus retained competitive absolute readout and improved retention over CK4 without becoming the unique retention optimum (Supplementary Tables S8 and S9).

Table 3. Compact main-method metrics. Paired cosine and L2 drift are means across 24 controlled WGS length-by-perturbation cells. Logistic macro-F1 is averaged across 132 fixed-split WGS readout cells and is a shallow accessibility probe rather than production-classifier performance. Feature count reports the fixed representation dimension.

representation	paired cosine	L2 drift	logistic macro-F1	features
CK4	0.969	0.220	0.363	136
CK4+P	0.984	0.157	0.364	147
CK4P-MSP	0.989	0.129	0.376	222
CK5	0.945	0.295	0.396	512

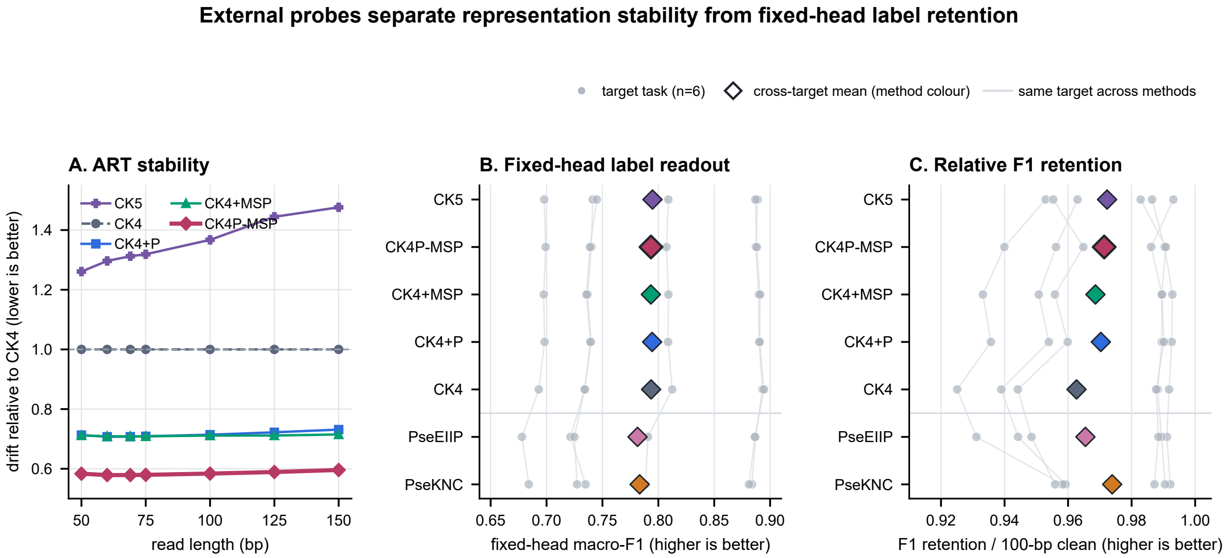

Figure 4 External probes separate simulator stability from fixed-head label retention. (A) Standardized L2 drift relative to CK4 at each ART read length; the horizontal CK4 reference is 1.0 and lower ratios indicate less within-length drift. The ART layer contains 55,961 matched clean/simulated pairs across 50, 60, 69, 75, 100, 125 and 150 bp. Lines are descriptive length-specific ratios without uncertainty intervals and are not interpreted as a monotonic physical length response. (B) Target-level mean macro-F1 across eight shifted CAMI_TOY_low length-by-condition cells. A class-balanced logistic head was trained only on 100-bp clean sources and reused without refitting. (C) Macro-F1 retention relative to each method's own 100-bp clean baseline. Panels B and C contain six prespecified one-versus-background tasks with 1,200 target and 1,200 background source groups each. Neutral circles denote target-task estimates, method-coloured diamonds denote cross-target means and faint lines preserve target pairing across methods. The six tasks share a CAMI source pool and are not independent-community replicates.

Alt text: ART drift ratios show lower relative drift for CK4P-MSP, while paired CAMI target-task panels show comparable absolute fixed-head readout among CK4-family methods and competitive, but not uniquely optimal, retention.

The quality-stratified ART audit and the CAMI II marine subset have different evidentiary roles. The former tests the current representation contract within simulator-derived read-length and quality strata; the latter applies controlled paired perturbations to anonymous public marine reads without reconstructing read-level taxonomic labels. Marine composition can differ materially from pathogen-rich or low-biomass mNGS sources, including in GC distribution and sequence complexity. The marine result therefore tests whether the stability ordering survives a composition-shifted public source; it is not external taxonomic validation.

Probe preprocessing created a separate boundary. In the original prespecified CAMI target task, CK4P-MSP retained mean shifted macro-F1 of 0.886 in contract space, with a worst shifted value of 0.881. Applying a coordinate-wise z-score fitted only on clean training reads reduced these values to 0.822 and 0.716. The largest N-mask-associated logit shifts occurred in low-variance hydrogen-bond and EIIP dispersion coordinates rather than in the explicit N-fraction coordinate. Across six targets, varying the contract-space logistic regularization over $C \in \{0.1, 1, 10\}$ changed the relative ordering of the CK4-family probes, whereas CK4P-MSP remained above PseKNC and PseEIIP at all three values. Low geometric drift therefore did not guarantee invariance to learned downstream preprocessing (Supplementary Figure S14).

Together, ART supports within-length simulator consistency, CAMI_TOY_low supports coarse fixed-head label retention, and CAMI II supports anonymous-read stability. The fixed-head tasks extend the controlled-grid observation to a public metagenomic source and several target definitions, while remaining a shared-source coarse-label audit rather than fine taxonomic validation.

The complete CAMI II marine length-by-perturbation matrix is provided in Supplementary Figure S10; fixed-head preprocessing and regularization sensitivities are reported in Supplementary Figure S14.

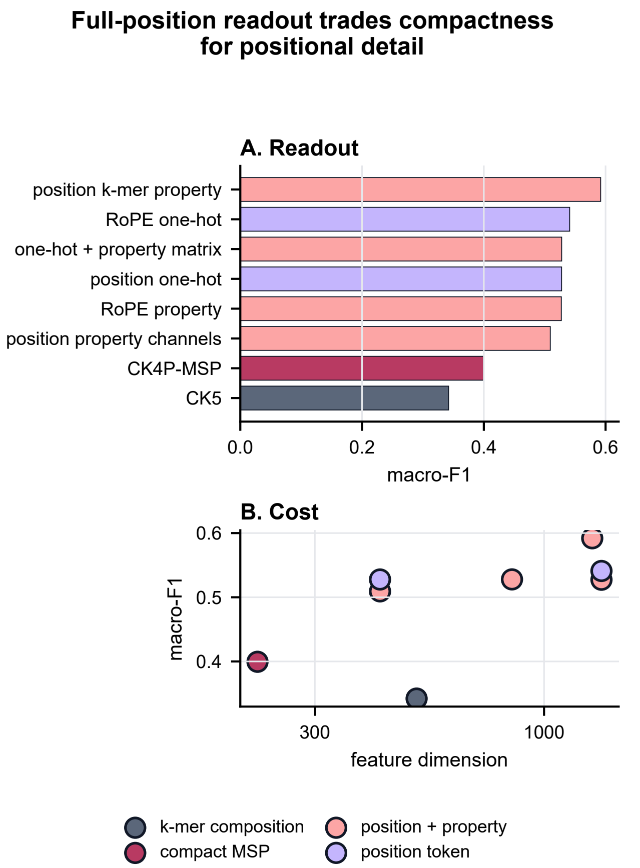

Figure 5 Full-position diagnostic upper bound for positional information. (A) Macro-F1 on controlled positional readout tasks compares compact summaries with representations that retain per-position token or property channels. (B) Readout is plotted against feature dimension to show the cost of preserving near-explicit positional information. CK4P-MSP provides a compact coarse summary and is not the positional-readout upper bound.

Alt text: Bar and scatter plots showing that full-position matrices add positional readout value at substantially higher feature dimension.

Full-position matrices define a positional upper bound rather than a deployment default

Full-position encodings were used to estimate what is gained when fine-grained layout information is retained. In controlled position and order tasks, full-position token or property matrices improved positional readability compared with compact summaries. This was expected, because these matrices preserve per-position structure that compact summaries intentionally compress.

The result is best interpreted as an upper-bound diagnostic. Full-position matrices clarify the value of positional information, but their high dimensionality and task specificity make them unsuitable as the default representation for compact short-read auditing. CK4P-MSP therefore occupies a different role: it does not match the full positional upper bound, but it retains a coarse positional property summary at a much smaller feature cost.

To test whether more explicit token-position coupling could strengthen this boundary without redefining the main method, we evaluated CK4P-MSP-PKM after freezing the CK4P-MSP contract. At $\delta = 0.25$, motif-position macro-F1 increased from 0.632 to 0.710 and grouped local-change macro-F1 from 0.888 to 0.928, whereas mean WGS drift increased from 0.129 to 0.138 and ART drift from 0.157 to 0.165. CAMI fixed-head retention was essentially unchanged (0.991 versus 0.993). The extension increased dimensionality from 222 to 297, median reference extraction time from 0.79 to 1.44 s per actual 10,000-read batch, and reverse-complement-pair drift from 0.078 to 0.251. A new-seed resampling check preserved the direction of the stability–readability exchange but reused the same benchmark resources and is not an independent external replication. In the common-contract historical profile, CK4P-MSP-PKM exceeded PseKNC and PseEIIP on the two structured readouts but did not exceed the length-dependent NCP+ANF positional control. The variant therefore exposes a selectable position–readability trade-off; it does not dominate CK4P-MSP or establish $\delta = 0.25$ as a general optimum (Supplementary Figure S15).

Local mutation analysis separates numerical stability from selective sensitivity

A stable audit representation should remain responsive to structured changes, but the relevant notion of responsiveness depends on the metric. We therefore tested whether local biochemical change remained distinguishable from matched nuisance perturbation. In a distance-ratio analysis, CK4P-MSP did not exceed CK4: its local-to-nuisance L2 ratio was 0.752 compared with 0.750 for CK4. Full-position and property-channel representations were stronger distance-amplification probes. CK4P-MSP's positive result was different: it retained high grouped delta-readout at compact dimensionality, with local-versus-nuisance macro-F1 of 0.979. The seven-group ablation assigned most of this local-readout increment to MSP, while P contributed primarily to global stability and K to nearest-clean retrieval.

This result supports a division of labor. Paired cosine and L2 drift quantify nuisance stability, distance ratios identify upper-bound local-change probes, and delta-readout asks whether structured local changes remain accessible to a shallow model. CK4P-MSP should therefore not be described as a distance-amplifying mutation detector. Its value in this analysis is that it remains stable under nuisance perturbation while preserving decomposable information for local-change readout at much lower dimensionality than full-position matrices.

The mutation-fraction sweep is shown in Supplementary Figure S4.

Boundary analyses constrain the interpretation

The boundary probes were included to prevent overextension of the representation claim. Spaced-seed transfer was useful as a mechanism comparator, but matching-oriented seed priors did not automatically become a universal dense-feature advantage. Context visibility depended on whether the relevant relation was

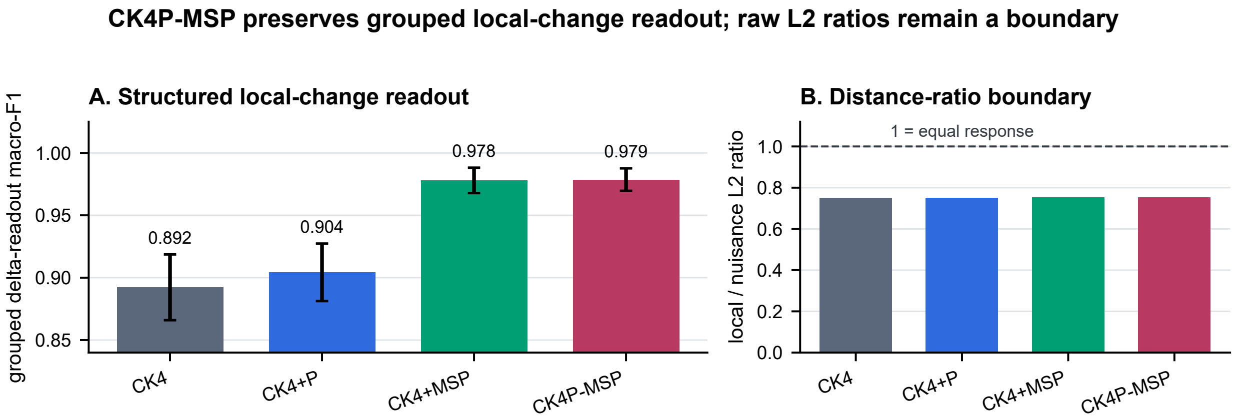

Figure 6 Local-change readout and distance-ratio boundary across 12 length-by-local-mode cells (250 template triplets per cell). (A) Mean local-versus-nuisance macro-F1 under template-grouped fivefold cross-validation shows that MSP accounts for most of the compact local-change readout increment. (B) Mean local-to-nuisance L2 ratio shows that CK4P-MSP does not exceed CK4 on raw distance amplification. Error bars are 95% matched-cell bootstrap intervals; they quantify variation across the controlled grid rather than independent biological-cohort uncertainty. Higher values indicate greater readout or distance selectivity, respectively.

Alt text: Bar plots showing MSP-driven grouped local-change readability and the separate compact-representation distance-ratio boundary.

Table 4. Seven-group block ablation. Global drift and nearest-clean retrieval are averaged over 24 matched length-by-perturbation cells. Grouped delta-readout macro-F1 is averaged over 12 length-by-local-mode cells using template-grouped fivefold cross-validation. Lower drift and higher retrieval/readout indicate stronger performance on their respective audit axes.

representation	features	L2 drift	retrieval	grouped macro-F1	observed audit role
CK4	136	0.218	1.000	0.892	local composition and composition-linked retrieval
P	11	0.025	0.295	0.639	global property stability without local-composition retrieval
MSP	75	0.027	0.958	0.965	coarse positional property readout
CK4+P	147	0.156	1.000	0.904	composition plus global property stability
CK4+MSP	211	0.156	1.000	0.978	composition plus local-change readability
P+MSP	86	0.027	0.942	0.970	stable property-only summary with incomplete retrieval
CK4P-MSP	222	0.129	1.000	0.979	balanced block-decomposable trade-off

Table 5. Boundary and mechanism summary for the spaced-seed, context-visibility and ARG/SNP probes. These analyses delimit claims that are not supported by the compact representation; they are mechanism and failure-boundary checks rather than additional performance benchmarks.

boundary	main observation	interpretation
spaced-seed transfer	The best tested 4-position dense stability pattern was contiguous 0-1-2-3 across the 69/75 bp sanity grid.	spaced seeds remain useful matching priors, but transfer is regime-specific
context visibility	motif-pair visibility thresholds depended on placement, emerging at 130, 140, 148 or 155 bp.	a representation cannot encode context absent from the read
ARG/SNP readout	stable compact features did not by themselves establish allele, resistance-SNP or functional equivalence.	identity/database evidence remains necessary for final biological calls

present within the read. ARG/SNP interpretation required identity, annotation and allele-level evidence beyond what compact property summaries can provide.

These negative and boundary results are part of the manuscript's logic. They keep the claim focused on representation diagnostics and representation-level perturbation auditing, rather than allowing the compact feature space to be misread as a full biological interpretation system.

Discussion

This study treats short-read sequence representation as an inspectable object rather than only as a prelude to classification. Database-linked exact matching remains distinct from the compact representation itself, whereas biochemical and coarse positional summaries change which perturbation responses can be resolved. CK4P-MSP occupies a compact trade-off in this setting: its reverse-complement canonical 4-mer composition block is augmented by biochemical and multi-scale property-pooling channels that reduce drift relative to CK4/CK5 while retaining local-change readability. Comparisons with historical descriptors place this contribution in a defined stability–readability regime rather than claiming first feature fusion or universal superiority.

The results support a division of labour between representation families. Canonical k-mer counts provide a compact local-composition backbone, while exact matching and alignment retain their distinct database-linked role. Within the K/P/MSP ablation, global biochemical summaries provided the strongest global stability channel. Multi-scale positional property pooling used the same biochemical property family to add coarse layout summaries to an otherwise order-poor k-mer-frequency vector and accounted for most of the grouped local-change readout gain. It is not a larger k-mer vocabulary, a learned token mapping or a reconstruction of full positional structure. Full-position matrices, by contrast, define what can be gained when positional information is preserved almost explicitly. They are useful diagnostic upper bounds but are not the preferred compact representation.

The exploratory CK4P-MSP-PKM extension further separates coarse layout from token-position coupling. Its stronger motif-position and local-change readouts show that explicit positional k-mer moments can recover information compressed by MSP, but the accompanying increases in drift, dimension, runtime and input-direction sensitivity prevent it from replacing the main CK4P-MSP contract. It is therefore a supplementary Pareto boundary rather than a second main method.

The additional analyses sharpen this interpretation. Same-dimension PCA/SVD controls show that numerical stability is not unique to CK4P-MSP: fitted CK7 projections achieved still lower drift, establishing a data-dependent reduction boundary. The historical audit reached the same conclusion from a different direction. PseKNC was both lower-dimensional and more stable than CK4P-MSP, but its grouped local-change readout was lower. NCP+ANF approached the CK4P-MSP local readout at a larger, length-dependent dimension and higher drift. CK4P-MSP therefore occupies an attributable stability–readability regime rather than the minimum-drift corner. Dimension-matched high-k vector controls show that this fixed representation remained more stable than the tested hashed and random-projection $k = 15$ controls at a comparable feature budget, although shallow readout differences were modest. MinHash was assessed separately in its native Jaccard geometry. Block-weight and fixed-map scaling audits show that the reported trade-off is not tied to one selected weight or reasonable fixed property remapping; the deterioration under train-fit coordinate z-scoring also shows that the geometry is not scale-invariant. Paired Wilcoxon tests show that the principal contrasts are consistent across the tested matched cells. The KSG-style audit indicates an estimator-dependent empirical separability increment beyond CK4 distance in the tested local-mutation regime.

The seven-group contribution audit makes this division of labour more precise. Property-only summaries were numerically stable but did not preserve the same nearest-clean retrieval as K-containing representations. Conditional on K+MSP, P reduced global drift without a detectable increment in the original local-versus-nuisance readout. Conditional on K+P, MSP reduced global drift and produced the largest grouped local-change readout increment. Conditional on P+MSP, K restored nearest-clean retrieval while increasing numerical drift. The factorial audit refined this assignment: MSP improved spatial-localization and chemistry readouts, whereas P added a smaller chemistry increment without a spatial increment. The redundancy audit further showed that P and MSP share a strong global property component but are not interchangeable: MSP has higher numerical rank and incomplete row-wise alignment with P. These results support metric-specific conditional contributions, not statistical independence, universal necessity or a claim that every block improves every endpoint.

The perturbation analyses also clarify the scope of the evidence. Uniform substitutions and local mutation sweeps provide controlled tests of representation behaviour. Across the shared-template 50–75 bp sweep, the CK4P-MSP stability–readability profile changed smoothly without a single-length discontinuity. The 100, 125 and 150 bp conditions remain broader anchors rather than part of a dense continuous sweep. ART adds current-contract Illumina-like simulator context: CK4P-MSP retained approximately 58–60% of CK4 drift within each tested length and preserved the same ordering across quality strata. Because the selected ART profile generated non-monotonic error loads across cycle lengths, these results support within-length consistency rather than a universal physical length law. CAMI_TOY_low adds six source-grouped one-versus-background tasks drawn from a shared source pool. CK4P-MSP preserved shifted macro-F1 comparable to CK4/CK5 and was descriptively above PseKNC/PseEIIP in each target definition, but it did not improve absolute readout over every internal block combination or maximize relative retention. The CAMI II marine subset provides a more complex anonymous-read stability probe without read-level taxonomic labels in this lightweight analysis. Together, these results support bounded external information retention across defined public-source probes, not independent-community, fine-taxonomic or clinical validation.

A useful outcome of the boundary analyses is that they prevent overgeneralization. CSP and spaced-seed controls help interpret which matching-oriented priors transfer into dense read-level representations, but they do not become general substitutes for compact composition-property features. Context and ARG/SNP probes show that compact stability is not the same as allele-level or functional interpretation. Those tasks require database identity, annotation, sufficient context and downstream biological evidence.

Limitations

The first limitation is task scope. The study evaluates representation diagnostics, whereas Kraken 2, Centrifuge, CLARK, Kaiju and related systems remain the appropriate reference frame for database-linked taxonomic assignment. The present work asks what information remains visible before or alongside such assignment systems. Direct integration with production classifiers, including downstream false-hit reduction or classifier-output auditing, remains future work and is not evaluated here.

The second limitation is the perturbation model and external benchmark scale. Controlled substitutions, ambiguous-base perturbations and local mutation sweeps are useful because they isolate representation behaviour, but they do not reproduce every physical process in sequencing data. ART adds current-contract simulator-derived context, although its cycle-specific error load prevents treating the seven ART lengths as a clean causal length experiment. CAMI_TOY_low and CAMI II marine add external metagenomic resources, but with different roles: the former supports six coarse fixed-head label tasks drawn from one source pool, and the latter supports only paired stability because read-level taxonomic labels were not reconstructed for the anonymous reads in this lightweight probe. The six CAMI target tasks are analysis blocks rather than independent communities, and source reads can recur as background across tasks. The resulting conclusions therefore remain bounded to controlled short-read, simulator-contextualized and external stability/readout settings, not to fine taxonomic assignment, production-scale community profiling or unobserved clinical error distributions.

The third limitation concerns read-length coverage. The dense shared-template audit covers the lower 50–75 bp range, while 100, 125 and 150 bp are discrete broader anchors. The study therefore does not estimate a continuous response over the intervening 75–150 bp interval. Individual lengths within the lower range are experimental conditions rather than biological or clinical thresholds.

The fourth limitation is that the MI and conditional-MI analyses are empirical. They support perturbation-associated signal in the tested feature channels, but they are not general evidence that a property-augmented representation always has higher mutual information than a composition-only representation. Likewise, mixed L2 distance is standardized representation drift after block construction and normalization, not a natural metric that equates biochemical properties and k-mer counts as identical physical units. Its magnitude depends on the declared property maps and block weights. Under the tested local-perturbation grid and estimator, property-aware distance summaries showed a positive empirical separability increment relative to composition-only and permutation controls; the weaker 150-bp cells delimit this observation.

The fifth limitation concerns MSP weighting and scale. The short-bin audit did not show an immediate loss of reliability across the 50–75 bp sweep, and finer multi-scale pooling increased delta-readout rather than immediately degrading it. Even so, the current MSP channel remains a static coarse summary. It is not an optimized variance-aware estimator, and the manuscript does not claim that its present binset or γ weighting is universally optimal across read lengths, perturbation types or sequencing platforms. The scaling audit further shows that train-fit variance amplification changes the geometry substantially; the reported method is therefore the declared encoding-value-weighted contract, not a scale-free construction.

The CK4P-MSP-PKM extension has a related but distinct limitation. Its fixed $\delta = 0.25$ value was retained from an exploratory grid and checked by new-seed resampling, not selected or validated on an independent biological benchmark. Odd positional moments also preserve input direction, so the assembled extension is not reverse-complement invariant as a whole. These constraints are reported alongside the readout gain and exclude a general superiority claim.

The sixth limitation is computational scope. After integer k-mer encoding and batched property pooling, the reference CK4P-MSP implementation required a median 2.71 s for an actual 10,000-read 75-bp batch, compared with 2.15 s for CK4 under the same protocol; a separate 100,000-read pass required 28.72 and 21.73 s, respectively. These are implementation-level measurements on one workstation, not hardware-independent complexity results. Their value in this manuscript is to bound the cost of interpretability and perturbation diagnostics under compact dimensionality rather than to claim engineering speed-up. Production-scale deployment would still require compiled or streaming implementations and pipeline-level cost-benefit testing.

The seventh limitation is descriptor, preprocessing and model scope. PseKNC, NCP+ANF and PseEIIP are representative historical controls rather than an exhaustive search over handcrafted descriptor families or their task-specific parameterizations; their deterministic ambiguous-base extensions are one declared implementation choice. The fixed-head audit also showed that coordinate-wise train-fit z-scoring can amplify low-variance P coordinates under N masking even when the declared contract-space geometry remains stable. Downstream scaling and regularization are therefore part of the readout contract, not innocuous implementation details. Large DNA language models and learned metagenomic classifiers provide powerful sequence representations and may outperform CK4P-MSP in predictive probes, especially when sufficient training data, pretraining and compute are available. CK4P-MSP instead provides a low-dimensional, training-free and block-decomposable coordinate system for auditing local composition, biochemical and coarse positional information under controlled short-read perturbation. Broader descriptor and learned-embedding comparisons would require a separate benchmark that reports predictive accuracy, preprocessing sensitivity and auditability together.

Future work

Future work should test whether CK4P-MSP or related compact summaries improve failure-mode detection when placed alongside database-centred pipelines. Pipeline-facing false-hit reduction or classifier-output auditing would require downstream classifier outputs and explicit operating-point analysis. A useful next step is to test whether composition-linked retrieval, biochemical stability and local-change readability can inform explicit representation-level warning signals.

A second direction is to expand the perturbation model. Platform-specific error-transition matrices, explicit Q-score trajectories and larger simulator grids would make the error-aware analysis more realistic. The current controlled simulations are useful because they isolate representation behaviour by read length, perturbation type and feature family; future work should test whether the same conclusions hold under richer platform-specific error processes. When ethically and legally feasible, restricted clinical read collections could then be used for external validation under controlled access, with public release limited to aggregate metadata and reproducible scripts.

A third direction is to make the positional channel more adaptive without changing the paper's current claim. Length-aware or variance-aware MSP weighting could be explored in future work, but such a step should be framed as a new model family rather than as a cosmetic parameter retune. The present results justify static MSP as a usable compact summary; they do not settle the separate optimization problem of how positional bins should be weighted under different noise and length conditions.

The CK4P-MSP-PKM audit also identifies a more specific extension problem: positional token moments could be made strand-aware by using paired even/odd summaries, explicit orientation pooling or reverse-complement symmetrization. Any such construction would require a new fixed contract and independent validation rather than retrospective adjustment of the reported $\delta = 0.25$ variant.

A fourth direction is to expand both historical and learned representation boundaries. Additional pseudo-composition, autocorrelation and task-specific hybrid descriptors should be evaluated under the same paired audit rather than assumed equivalent from their feature definitions. Full-position matrices and DNA foundation-model embeddings are also natural candidates because they may recover richer context and stronger predictive signal than compact summaries. The present work suggests where such models should be audited: not only by final classification accuracy, but also by composition retention, perturbation stability, positional readability and local-change sensitivity.

Conclusion

This study shows that compact short-read representations can be evaluated as interpretable audit objects. In controlled settings, CK4P-MSP combined a canonical local k-mer-composition block with global and positionalized biochemical channels to reduce perturbation drift relative to CK4/CK5 while preserving grouped local-change readability. PseKNC established a lower-drift, lower-dimensional historical boundary, whereas CK4P-MSP retained stronger structured local-change readout and explicit K/P/MSP attribution. Across six source-grouped external coarse-label tasks, CK4P-MSP retained fixed-head readout comparable to CK4/CK5 and was descriptively above PseKNC/PseEIIP; downstream scaling exposed a distinct N-mask failure mode. Compact biochemical and coarse position-aware summaries can therefore provide block-decomposable coordinates for representation-level perturbation and preprocessing audits alongside database-linked exact matching and curated reference methods.

Data Availability

All processed data tables, figure source summaries and analysis outputs generated for this study are maintained in the versioned GitHub repository ultrashort-dna-representation-diagnostics, on the `release` branch. The frozen submission snapshot is GitHub Release `v1.1.3`; all archived versions are linked by the Zenodo concept DOI doi:10.5281/zenodo.21792340.

Publicly reused resources include simulator- and benchmark-derived materials used for the shared-template WGS continuity audit, ART, CAMI_TOY_low, CAMI II marine and Dorsal motif-position analyses [52, 14, 15, 53]. The public release contains the 420-template continuity manifest, compressed perturbation summaries, local-triplet results, ART generation manifests, quality-stratified summaries and the small public motif-position source package. The CAMI II marine lightweight probe used the public CAMI II marine short-read sample 0 reads archive and its corresponding setup archive; exact resource URLs are listed in the release repository. Only a streamed prefix was parsed for the lightweight stability probe, and read-level taxonomic labels were not reconstructed from the separate truth resources for this analysis.

Code Availability

The analysis scripts used to generate tables, figures and methodological audits are maintained in the same MIT-licensed repository. The repository README and release manifest map each manuscript figure, table and supplementary audit to its source script, input tables and generated outputs. The exact software snapshot used for submission is available as GitHub Release `v1.1.3`; the corresponding Zenodo version record is discoverable through doi:10.5281/zenodo.21792340.

Supplementary Data

Supplementary Data are available at *NAR Genomics and Bioinformatics* Online. The supplementary file includes Supplementary Figures S1–S15 and Supplementary Tables S1–S9. The underlying machine-readable source tables and figure-generation inputs are included in the public archived release.

Ethics and Data Governance

This study used only publicly available reference and benchmark data and computationally simulated sequences. No human participants, patient-level data or identifiable private information were involved; therefore, institutional ethics approval was not required.

Author Contributions

Ruixiang Mei: Conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing – original draft, and writing – review and editing. ORCID: `https://orcid.org/0009-0003-2128-0726`.

AI-Assisted Tools Disclosure

Generative-AI tools (OpenAI ChatGPT and Codex, Google Gemini, and Doubao) were used primarily as assistive review and quality-control tools for language, terminology, internal consistency, references, formatting and code, including support for preparing and checking plotting scripts. No AI-generated output was used as experimental data or treated as scientific evidence. Scientific decisions, interpretation of the results and final wording remained under the author's control. The author independently reviewed, revised and verified all manuscript text, code, calculations, references and figures and assumes full responsibility for the work.

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Conflict of Interest

The author declares no competing interests.

References

1. Michael R. Wilson, Samia N. Naccache, Erika Samayoa, Mark Biagtan, Ali Bashir, Guixia Yu, S. M. Salamat, Sneha Somasekar, Scot Federman, Steve Miller, Robert Sokolic, Elitza Garabedian, Fabio Candotti,

Rebecca H. Buckley, Kurt D. Reed, Terry L. Meyer, Christine M. Seroogy, Renee Galloway, Stuart L. Henderson, James E. Gern, Joseph L. DeRisi, and Charles Y. Chiu. Actionable diagnosis of neuroleptospirosis by next-generation sequencing. *New England Journal of Medicine*, 370:2408–2417, 2014. doi: 10.1056/NEJMoa1401268.

2. Michael R. Wilson, Heather A. Sample, Kaitlyn C. Zorn, Samuel Arevalo, Guixia Yu, John Neuhaus, Scot Federman, Deborah Stryke, Brian Briggs, Charles Langelier, Adam Berger, Victoria Douglas, S. Andrew Josephson, Felicia C. Chow, Blair D. Fulton, Joseph L. DeRisi, Jeffrey M. Gelfand, Samia N. Naccache, Jeffrey M. Bender, and Charles Y. Chiu. Clinical metagenomic sequencing for diagnosis of meningitis and encephalitis. *New England Journal of Medicine*, 380: 2327–2340, 2019. doi: 10.1056/NEJMoa1803396.
3. Charles Y. Chiu and Steven A. Miller. Clinical metagenomics. *Nature Reviews Genetics*, 20:341–355, 2019. doi: 10.1038/s41576-019-0113-7.
4. Marcel Martin. Cutadapt removes adapter sequences from high-throughput sequencing reads. *EMBnet.journal*, 17:10–12, 2011. doi: 10.14806/ej.17.1.200.
5. Anthony M. Bolger, Marc Lohse, and Bjoern Usadel. Trimmomatic: A flexible trimmer for illumina sequence data. *Bioinformatics*, 30: 2114–2120, 2014. doi: 10.1093/bioinformatics/btu170.
6. Susannah J. Salter, Michael J. Cox, Elina M. Turek, Szymon T. Calus, William O. Cookson, Miriam F. Moffatt, Paul Turner, Julian Parkhill, Nicholas J. Loman, and Alan W. Walker. Reagent and laboratory contamination can critically impact sequence-based microbiome analyses. *BMC Biology*, 12:87, 2014. doi: 10.1186/s12915-014-0087-z.
7. K. Eric Wommack, Jaysheel Bhavsar, and Jacques Ravel. Metagenomics: read length matters. *Applied and Environmental Microbiology*, 74(5):1453–1463, 2008. doi: 10.1128/AEM.02181-07.
8. William S. Pearman, Nikki E. Freed, and Olin K. Silander. Testing the advantages and disadvantages of short- and long-read eukaryotic metagenomics using simulated reads. *BMC Bioinformatics*, 21(1):220, 2020. doi: 10.1186/s12859-020-3528-4.
9. Derrick E. Wood and Steven L. Salzberg. Kraken: Ultrafast metagenomic sequence classification using exact alignments. *Genome Biology*, 15:R46, 2014. doi: 10.1186/gb-2014-15-3-r46.
10. Peter Menzel, Kim Lee Ng, and Anders Krogh. Fast and sensitive taxonomic classification for metagenomics with kaiju. *Nature Communications*, 7:11257, 2016. doi: 10.1038/ncomms11257.
11. Derrick E. Wood, Jennifer Lu, and Ben Langmead. Improved metagenomic analysis with kraken 2. *Genome Biology*, 20:257, 2019. doi: 10.1186/s13059-019-1891-0.
12. Rachid Ounit, Steve Wanamaker, Timothy J. Close, and Stefano Lonardi. Clark: Fast and accurate classification of metagenomic and genomic sequences using discriminative k-mers. *BMC Genomics*, 16: 236, 2015. doi: 10.1186/s12864-015-1419-2.
13. Daehwan Kim, Li Song, Florian P. Breitwieser, and Steven L. Salzberg. Centrifuge: Rapid and sensitive classification of metagenomic sequences. *Genome Research*, 26:1721–1729, 2016. doi: 10.1101/gr.210641.116.
14. Alexander Sczyrba, Peter Hofmann, Peter Belmann, David Koslicki, Stefan Janssen, Johannes Droege, Ivan Gregor, Stephan Majda, Jessika Fiedler, Eik Dahms, Andreas Bremges, Adrian Fritz, Ruben Garrido-Oter, Tue Sparholt Jorgensen, Nicole Shapiro, Philip D. Blood, Alexey Gurevich, Yang Bai, Dmitrij Turaev, Matthew Z. DeMaere, Rayan Chikhi, Niranjan Nagarajan, Christopher Quince, Fernando Meyer, Monika Balvociute, Lars Hestbjerg Hansen, Soren J. Sorensen, Nicholas Chia, Bertrand Denis, Jeff L. Froula, Zhong Wang, Rob Egan, Dongwan Don Kang, Jeffrey J. Cook, Charles Deltel, Michael Beckstette, Claire Lemaitre, Pierre Peterlongo, Guillaume Rizk, Dominique Lavenier, Yu-Wei Wu, Steven W. Singer, Chirag Jain, Marc Strous, Heiner Klingenberg, Peter Meinicke, Michael D. Barton, Thomas Lingner, Hsin-Hung Lin, Yu-Chieh Liao, Genivaldo Gueiros Z. Silva, Daniel A. Cuevas, Robert A. Edwards, Surya Saha, Vitor C. Piro, Bernhard Y. Renard, Mihai Pop, Hans-Peter Klenk, Markus Goeker, Nikos C. Kyrpides, Tanja Woyke, Julia A. Vorholt, Paul Schulze-Lefert, Edward M. Rubin, Aaron E. Darling, Thomas Rattei, and Alice C. McHardy. Critical assessment of metagenome interpretation - a benchmark of metagenomics software. *Nature Methods*, 14: 1063–1071, 2017. doi: 10.1038/nmeth.4458.
15. Fernando Meyer, Adrian Fritz, Zhi-Luo Deng, David Koslicki, Alexey Gurevich, Gary Robertson, Mohammed Alser, Dmitry Antipov, Francesco Beghini, Denis Bertrand, Jaqueline J. Brito, C. Titus Brown, Jan Buchmann, Aydin Buluc, Bo Chen, Rayan Chikhi, Philip T. L. C. Clausen, Alesia Cristian, Piotr W. Dabrowski, Aaron E. Darling, Rob Egan, Eleazar Eskin, Evangelos Georganas, Eugene Goltsman, Melissa A. Gray, Lars Hestbjerg Hansen, Steven Hofmeyr, Pingqin Huang, Luiz Irber, Hongying Jia, Tue Sparholt Jorgensen, Md. Rezaul Karim, Terje Klemetsen, Axel Kola, Sergey Koren, Jason Kwan, Nathan LaPierre, Claire Lemaitre, Chen Li, Antoine Limasset, Fabio Malcher-Miranda, Serghei Mangul, Vanessa R. Marcelino, Camille Marchet, Pierre Marijon, Dmitry Meleshko, Daniel R. Mende, Alessio Milanese, Niranjan Nagarajan, Jakob Nissen, Sergey Nurk, Leonid Oliker, David Paez-Espino, Pierre Peterlongo, Vitor C. Piro, Jacob S. Porter, Simon Rasmussen, Evan R. Rees, Knut Reinert, Bernhard Y. Renard, Espen M. Robertsen, Gail L. Rosen, Hans-Joachim Ruscheweyh, Varuni Sarwal, Nicola Segata, Enrico Seiler, Lizhen Shi, Fengzhu Sun, Shinichi Sunagawa, Soren J. Sorensen, Torsten Thomas, Chengchen Tong, Mirko Trajkovski, Julien Tremblay, Gherman V. Uritskiy, Riccardo Vicedomini, Zhong Wang, Yuzhen Ye, Pelin Yilmaz, Ronghui You, Georg Zeller, Sen Zhao, Shanfeng Zhu, Shaochun Zhu, Ruben Garrido-Oter, Petra Gastmeier, Stephane Hacquard, Susanne Haussler, Ariane Khaledi, Friederike Maechler, Fantin Mesny, Simona Radutoiu, Paul Schulze-Lefert, Nathiana Smit, Till Strowig, Andreas Bremges, Alexander Sczyrba, and Alice C. McHardy. Critical assessment of metagenome interpretation: the second round of challenges. *Nature Methods*, 19:429–440, 2022. doi: 10.1038/s41592-022-01431-4.
16. Karel Brinda, Michal Sykulski, and Gregory Kucherov. Spaced seeds improve k-mer-based metagenomic classification. *Bioinformatics*, 31: 3584–3592, 2015. doi: 10.1093/bioinformatics/btv419.
17. Pawel S. Krawczyk, Leszek Lipinski, and Andrzej Dziembowski. Plasflow: Predicting plasmid sequences in metagenomic data using genome signatures. *Nucleic Acids Research*, 46:e35, 2018. doi: 10.1093/nar/gkx1321.
18. Antonino Fiannaca, Laura La Paglia, Massimo La Rosa, Giosue' Lo Bosco, Antonio Urso, et al. Deep learning models for bacteria taxonomic classification of metagenomic data. *BMC Bioinformatics*, 19(Suppl 7):198, 2018. doi: 10.1186/s12859-018-2182-6.
19. Wolfgang Fuhl, Susanne Zabel, and Kay Nieselt. Resource saving taxonomy classification with k-mer distributions and machine learning, 2023.
20. Brian D. Ondov, Todd J. Treangen, Pall Melsted, Adam B. Mallonee, Nicholas H. Bergman, Sergey Koren, and Adam M. Phillippy. Mash: Fast genome and metagenome distance estimation using minhash. *Genome Biology*, 17:132, 2016. doi: 10.1186/s13059-016-0997-x.
21. Andrzej Zielezinski, Susana Vinga, Jonas Almeida, and Wojciech M. Karlowski. Alignment-free sequence comparison: Benefits, applications, and tools. *Genome Biology*, 18:186, 2017. doi: 10.1186/s13059-017-1319-7.
22. Hong Yi, Yao Lin, Wei Jin, et al. Sequences dimensionality-reduction by k-mer substring space sampling enables effective resemblance- and containment-analysis for large-scale omics-data. *Genome Biology*, 22: 84, 2021. doi: 10.1186/s13059-021-02303-4.
23. Will P. M. Rowe, Anna Paola Carrieri, Cristina Alcon-Giner, Shabhonam Caim, Alexander Shaw, Kevin Sim, J. Simon Kroll, Louise J. Hall, Edward O. Pyzer-Knapp, and Maureen D. Winn. Streaming histogram sketching for rapid microbiome analytics. *Microbiome*, 7: 40, 2019. doi: 10.1186/s40168-019-0653-2.
24. H. Joel Jeffrey. Chaos game representation of gene structure. *Nucleic Acids Research*, 18:2163–2170, 1990. doi: 10.1093/nar/18.8.2163.
25. Richard F. Voss. Evolution of long-range fractal correlations and 1/f noise in dna base sequences. *Physical Review Letters*, 68:3805–3808, 1992. doi: 10.1103/PhysRevLett.68.3805.
26. Dimitris Anastassiou. Genomic signal processing. *IEEE Signal Processing Magazine*, 18:8–20, 2001. doi: 10.1109/79.939833.
27. Paul D. Cristea. Conversion of nucleotides sequences into genomic signals. *Journal of Cellular and Molecular Medicine*, 6:279–303, 2002. doi: 10.1111/j.1582-4934.2002.tb00196.x.
28. Gerardo Mendizabal-Ruiz, Israel Román-Godínez, Sulema Torres-Ramos, Ricardo A. Salido-Ruiz, and J. Alejandro Morales. On dna numerical representations for genomic similarity computation. *PLOS ONE*, 12(3):e0173288, 2017. doi: 10.1371/journal.pone.0173288.
29. Gerardo Mendizabal-Ruiz, Israel Román-Godínez, Sulema Torres-Ramos, Ricardo A. Salido-Ruiz, and J. Alejandro Morales. Genomic signal processing for dna sequence clustering. *PeerJ*, 6:e4264, 2018. doi: 10.7717/peerj.4264.
30. Achuthsankar S. Nair and Sivarama Pillai Sreenadhan. A coding measure scheme employing electron-ion interaction pseudopotential. *Bioinformation*, 1:197–202, 2006.
31. Wei Chen, Hui Yang, Pengmian Feng, Hui Ding, and Hao Lin. iDNA4mC: identifying DNA N4-methylcytosine sites based on nucleotide chemical properties. *Bioinformatics*, 33(22):3518–3523, 2017. doi: 10.1093/bioinformatics/btx479.
32. Wei Chen, Tian-Yu Lei, Dian-Chuan Jin, Hao Lin, and Kuo-Chen Chou. PseKNC: A flexible web server for generating pseudo k-tuple nucleotide composition. *Analytical Biochemistry*, 456:53–60, 2014. doi: 10.1016/j.ab.2014.04.001.
33. Wei Chen, Xitong Zhang, Jordan Brooker, Hao Lin, Liqing Zhang, and Kuo-Chen Chou. PseKNC-General: a cross-platform package for generating various modes of pseudo nucleotide compositions. *Bioinformatics*, 31(1):119–120, 2015. doi: 10.1093/bioinformatics/btu602.
34. Bin Liu, Fule Liu, Longyun Fang, Xiaolong Wang, and Kuo-Chen Chou. repDNA: a python package to generate various modes of feature vectors for DNA sequences by incorporating user-defined physicochemical properties and sequence-order effects. *Bioinformatics*, 31(8): 1307–1309, 2015. doi: 10.1093/bioinformatics/btu820.

35. Wenying He, Cangzhi Jia, Yucong Duan, and Quan Zou. 70ProPred: a predictor for discovering sigma70 promoters based on combining multiple features. *BMC Systems Biology*, 12(Suppl 4):44, 2018. doi: 10.1186/s12918-018-0570-1.
36. Zhen Chen, Xuhan Liu, Pei Zhao, Chen Li, Yanan Wang, Fuyi Li, Tatsuya Akutsu, Chris Bain, Robin B. Gasser, Junzhou Li, Zuoren Yang, Xin Gao, Lukasz Kurgan, and Jiangning Song. iFeatureOmega: an integrative platform for engineering, visualization and analysis of features from molecular sequences, structural and ligand data sets. *Nucleic Acids Research*, 50(W1):W434–W447, 2022. doi: 10.1093/nar/gkac351.
37. Saman Zabihi, Sattar Hashemi, and Eghbal Mansoori. EDEN: multiscale expected density of nucleotide encoding for enhanced DNA sequence classification with hybrid deep learning. *BMC Bioinformatics*, 27(1):40, 2026. doi: 10.1186/s12859-026-06367-6.
38. Qiaoxing Liang, Paul W. Bible, Youping Liu, Bin Zou, and Li Wei. Deepmicrobes: Taxonomic classification for metagenomics with deep learning. *NAR Genomics and Bioinformatics*, 2:lqaa009, 2020. doi: 10.1093/nargab/lqaa009.
39. Felix Wichmann, Jose R. Zamudio, Roland Eils, and Matthias Schlesner. Metatransformer: Deep metagenomic sequencing read classification using self-attention models. *NAR Genomics and Bioinformatics*, 5:lqad082, 2023. doi: 10.1093/nargab/lqad082.
40. Florian Mock, Fleming Kretschmer, Anton Kriese, Sebastian Böcker, and Manja Marz. Taxonomic classification of dna sequences beyond sequence similarity using deep neural networks. *Proceedings of the National Academy of Sciences*, 119(35):e2122636119, 2022. doi: 10.1073/pnas.2122636119.
41. Pablo Millan Arias, Niousha Sadjadi, Monireh Safari, ZeMing Gong, Austin T. Wang, Joakim Bruslund Haurum, Iuliia Zarubiieva, Dirk Steinke, Lila Kari, Angel X. Chang, Scott C. Lowe, and Graham W. Taylor. Barcodebert: Transformers for biodiversity analyses. *Bioinformatics Advances*, 6(1):vbag054, 2026. doi: 10.1093/bioadv/vbag054.
42. Zhihan Zhou, Yanrong Ji, Weijian Li, Pratik Dutta, Ramana V. Davuluri, and Han Liu. Dnabert-2: Efficient foundation model and benchmark for multi-species genomes. In *International Conference on Learning Representations*, 2024. URL `https://openreview.net/forum?id=oMLQB4EZE1`.
43. Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam H. Grzywaczewski, Francesco Oteri, Christian Dallago, Evan Trop, Hassan Sirelkhatim, Guillaume Richard, Marcin J. Skwark, Karim Beguir, Monica Lopez, and Thomas Pierrot. Nucleotide transformer: Building and evaluating robust foundation models for human genomics. *Nature Methods*, 22:287–297, 2025. doi: 10.1038/s41592-024-02523-z.
44. Eric Nguyen, Michael Poli, Marjan Faizi, Armin W. Thomas, Camden Birch-Sykes, Michael Wornow, Aman Patel, Charles Rabideau, Stefano Massaroli, Yoshua Bengio, Stefano Ermon, Stephen A. Baccus, and Christopher Re. Hyenadna: Long-range genomic sequence modeling at single nucleotide resolution. In *Advances in Neural Information Processing Systems*, volume 36, pages 43177–43201, 2023. doi: 10.52202/075280-1872.
45. Eric Nguyen, Michael Poli, Matthew G. Durrant, Brian Kang, Dhruva Katrekar, David B. Li, Liam J. Bartie, Armin W. Thomas, Samuel H. King, Garyk Brixi, Jeremy Sullivan, Madelena Y. Ng, Ashley Lewis, Aaron Lou, Stefano Ermon, Stephen A. Baccus, Tina Hernandez-Boussard, Christopher Re, Patrick D. Hsu, and Brian L. Hie. Sequence modeling and design from molecular to genome scale with evo. *Science*, 386:eado9336, 2024. doi: 10.1126/science.ado9336.
46. Veniamin Fishman, Yuri Kuratov, Aleksei Shmelev, Maxim Petrov, Dmitry Penzar, Denis Shepelin, Nikolay Chekanov, Olga Kardymon, and Mikhail Burtsev. Gena-lm: A family of open-source foundational dna language models for long sequences. *Nucleic Acids Research*, 53: gkae1310, 2025. doi: 10.1093/nar/gkae1310.
47. Jie Ren, Kai Song, Chao Deng, Nathan A. Ahlgren, Jed A. Fuhrman, Yi Li, Xiaohui Xie, Ryan Poplin, and Fengzhu Sun. Identifying viruses from metagenomic data using deep learning. *Quantitative Biology*, 8 (1):64–77, 2020. doi: 10.1007/s40484-019-0187-4.
48. Yan Miao, Fu Liu, Tao Hou, and Yun Liu. Virtifier: A deep learning-based identifier for viral sequences from metagenomes. *Bioinformatics*, 38(5):1216–1222, 2022. doi: 10.1093/bioinformatics/btab845.
49. Yan Miao, Jilong Bian, Guanghui Dong, and Tianhong Dai. Detire: A hybrid deep learning model for identifying viral sequences from metagenomes. *Frontiers in Microbiology*, 14:1169791, 2023. doi: 10.3389/fmicb.2023.1169791.
50. Matthew P. Moore, Mirjam Laager, Paolo Ribeca, and Xavier Didelot. Kmeraperture: Retaining k-mer synteny for alignment-free extraction of core and accessory differences between bacterial genomes. *PLOS Genetics*, 20(4):e1011184, 2024. doi: 10.1371/journal.pgen.1011184.
51. Qiwen Yang, editor. *Practice and Progress of mNGS Report Interpretation*. Shanghai Science and Technology Press, 2026. Chinese-language edition; read-length context cited from p. 99.
52. Weichun Huang, Leping Li, Jason R. Myers, and Gabor T. Marth. Art: A next-generation sequencing read simulator. *Bioinformatics*, 28: 593–594, 2012. doi: 10.1093/bioinformatics/btr708.
53. Jacob Clifford and Christoph Adami. Data from: Discovery and information-theoretic characterization of transcription factor binding sites that act cooperatively, 2016. URL `https://doi.org/10.5061/dryad.8b203`. Dataset.
54. Conrad L. Schoch, Stacy Ciufo, Mikhail Domrachev, Carol L. Hotton, Sivakumar Kannan, Rogneda Khovanskaya, Detlef Leipe, Richard McVeigh, Kathleen O'Neill, Barbara Robbertse, Shobha Sharma, Vladimir Soussov, John P. Sullivan, Lu Sun, Seán Turner, and Ilene Karsch-Mizrachi. NCBI Taxonomy: a comprehensive update on curation, resources and tools. *Database*, 2020:baaa062, 2020. doi: 10.1093/database/baaa062.
55. Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945. doi: 10.2307/3001968.
56. Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1): 289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.
57. Alexander Kraskov, Harald Stögbauer, and Peter Grassberger. Estimating mutual information. *Physical Review E*, 69(6):066138, 2004. doi: 10.1103/PhysRevE.69.066138.
58. Brian C. Ross. Mutual information between discrete and continuous data sets. *PLOS ONE*, 9(2):e87357, 2014. doi: 10.1371/journal.pone.0087357.